Name:____________

AS Physics Unit 2

Once you have completed a paper and have marked it, fill out the table below, using the grade boundaries provided to add your grade. Then list which questions you did well and which questions you struggled with. For the questions your struggled with, review why and at the reasons. The most common reasons are

- Not understanding basic concepts (e.g. simple definitions)

- Calculation errors (forgetting to square numbers, etc.)

- Not reading questions carefully

- Not answering the full question (describe vs. explain)

- Laboratory/practical setup mistakes

For these questions it is also useful add notes to the question to help you next time and take a photo of these questions. Use these then to guide further revision.

Year	Score /80	Grade	Good Questions	Bad Questions	Reasons
2016					
2017					
2018					
2019					
2022					
2023					

Grade boundaries - Raw Score

Year	E	D	C	B	A	Max UMS
2016	19	26	33	40	47	61
2017	18	25	33	41	49	65
2018	17	24	31	38	45	59
2019	20	27	34	41	49	65
2022	10	17	25	33	41	57
2023	15	22	29	37	45	61

Surname	Centre Number	Candidate Number
Other Names		2

GCE AS/A Level

2420U20-1 – **NEW AS**

S16-2420U20-1

PHYSICS – Unit 2
Electricity and Light

P.M. THURSDAY, 9 June 2016

1 hour 30 minutes

For Examiner's use only		
Question	**Maximum Mark**	**Mark Awarded**
1.	**8**	
2.	**13**	
3.	**9**	
4.	**9**	
5.	**14**	
6.	**10**	
7.	**17**	
Total	**80**	

2420U201 01

ADDITIONAL MATERIALS

In addition to this paper, you will require a calculator and a **Data Booklet**.

INSTRUCTIONS TO CANDIDATES

Use black ink or black ball-point pen. Do not use pencil or gel pen. Do not use correction fluid.

Write your name, centre number and candidate number in the spaces at the top of this page.

Answer **all** questions.

Write your answers in the spaces provided in this booklet. If you run out of space use the continuation pages at the back of the booklet taking care to number the question(s) correctly.

INFORMATION FOR CANDIDATES

The total number of marks available for this paper is 80.

The number of marks is given in brackets at the end of each question or part-question.

You are reminded to show all working. Credit is given for correct working even when the final answer is incorrect.

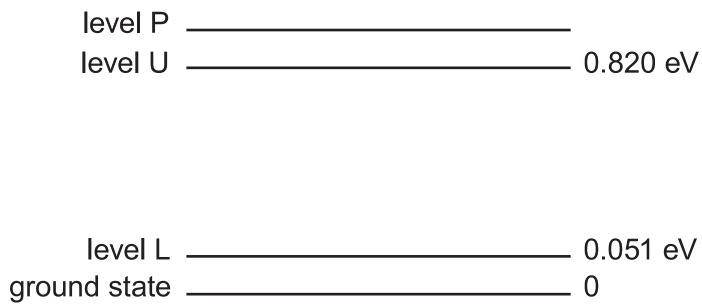
The assessment of the quality of extended response (QER) will take place in **Q7*(b)*(ii)**.

JUN162420U20101

Examiner only

Answer **all** *questions*

1. A simplified energy level system is given for a particular four-level laser system. Electrons are pumped (by means of infra-red radiation) from the ground state to level P, and drop to U, setting up a population inversion.



(a) (i) Calculate the wavelength of radiation emitted in the transition from level **U** to level **L**. [3]

(ii) Explain how stimulated emission enables amplification of infra-red radiation of this wavelength. [3]

(b) Explain the advantage of a four-level laser system over a three-level system. [2]

2420U201
03

Examiner only

2. *(a)* (i) Define the *work function* of a material. [1]

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(ii) When a potassium surface is irradiated with light of frequency 7.4×10^{14} Hz, electrons of maximum kinetic energy 1.2×10^{-19} J are ejected at a rate of 2.0×10^{15} electrons per second.

I. Explain, in terms of photons, how, if at all, the maximum kinetic energy of the ejected electrons **and** their rate of ejection would change if a more intense light of the same frequency were used. [3]

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II. Determine whether or not electrons would be ejected from a potassium surface by light of frequency 5.1×10^{14} Hz. Give your reasoning. [4]

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Examiner only

(b) A beam of monochromatic light of wavelength, λ and power, P strikes an absorbing surface normally.

(i) Derive an expression for the number of photons, N, striking the surface per second, in terms of P, λ, h and c. [2]

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(ii) Hence derive an expression for the momentum change per second of the light when it strikes the surface. [2]

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(iii) A student suggests that the answer to (ii) gives the *pressure* that the light exerts on the surface. What *should* she have said instead of *pressure*? [1]

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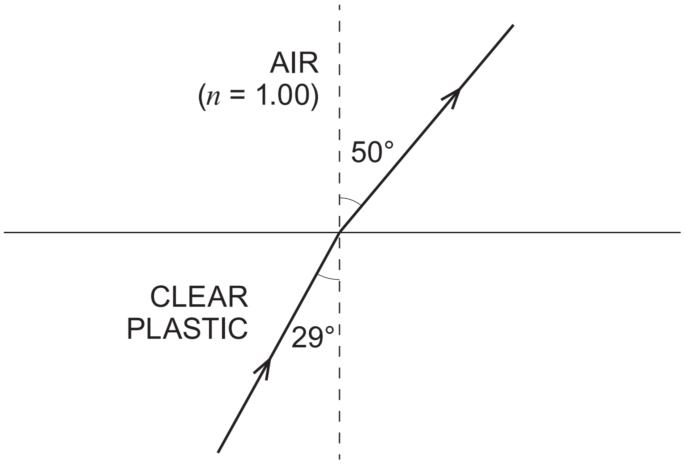
2420U201 05

Examiner only

3. *(a)* A narrow beam of light is observed to refract as shown between clear plastic and air.



(i) Calculate the speed of light in the **plastic**. [2]

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(ii) Calculate the critical angle for light approaching air from the plastic. [1]

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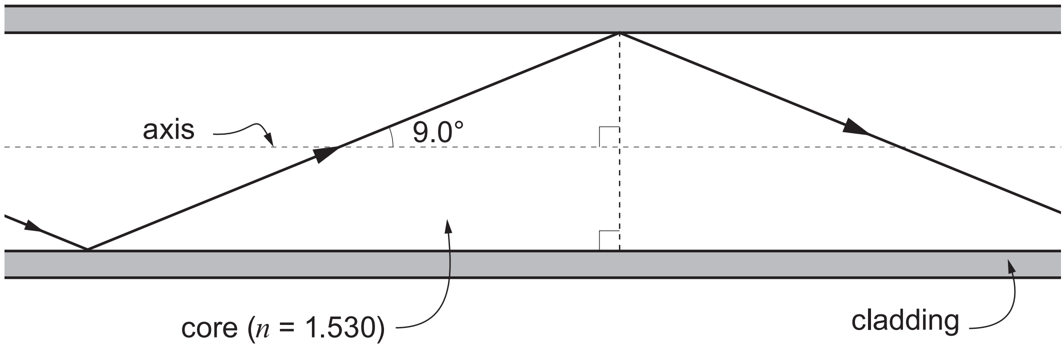
Examiner only

(b) The diagram shows light being transmitted along a multimode fibre, at the greatest angle to the axis for successful transmission.



NOT TO SCALE

(i) The refractive index of the core is 1.530. Calculate the refractive index of the cladding. [2]

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(ii) Show that the zigzag route in the diagram is 1.0125 times longer than a straight path through the same length of fibre. [1]

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(iii) The difference in times of travel for a data pulse by these two extreme routes is required to be no more than 7.5 ns. Determine whether or not 150 m of fibre will be too long. Set out your reasoning clearly. [3]

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2420U201 07

Examiner only

4. A beam of monochromatic light is shone normally (at right angles) on to a diffraction grating.

(a) **Explain** in clear steps why bright beams emerge from the grating at angles, θ, to the normal given by the equation:

$$d \sin \theta = n\lambda$$ [3]

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(b) The angles, θ, at which the bright beams emerge are given in the table below.

Order, n	**θ (mean)**	
0	0	
1	16	
2	35	
3	58	

(i) Plot a graph of sin θ (y-axis) against n (x-axis) on the grid provided. [3]

Examiner only

2420U201 09

(ii) **Use your graph** to determine the wavelength of the light. The separation between the centres of slits in the grating is 1800 nm. Show your working. [3]

Examiner only

5. *(a)* Define the *potential difference* between two points in an electric circuit. [2]

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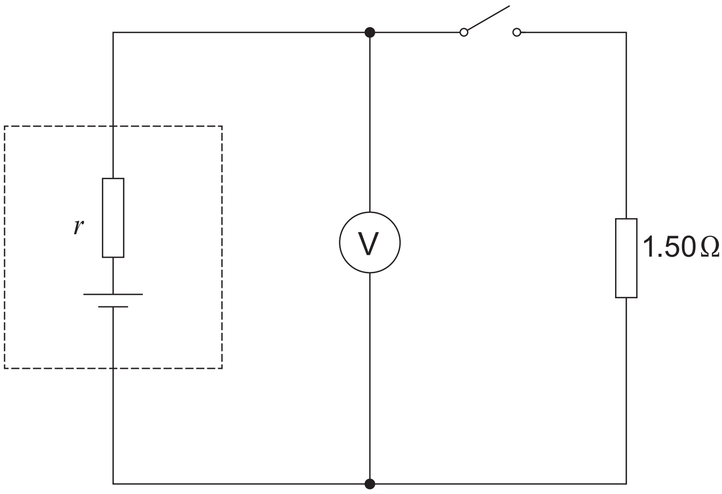
(b) A cell of emf 1.62 V and internal resistance, r, is included in the circuit shown.



(i) State the expected reading on the voltmeter when the switch is open. .... [1]

(ii) With the switch closed the voltmeter reads 1.38 V. Show in clear steps that the cell's internal resistance, r, is approximately 0.3 Ω. [3]

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(iii) 750 J of the cell's energy is dissipated **in total** while the switch is closed. Calculate the time for which the switch is closed. [2]

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Examiner only

(iv) Calculate the voltmeter reading when the 1.50 Ω resistor is replaced by a 0.75 Ω resistor, and the switch is closed. [2]

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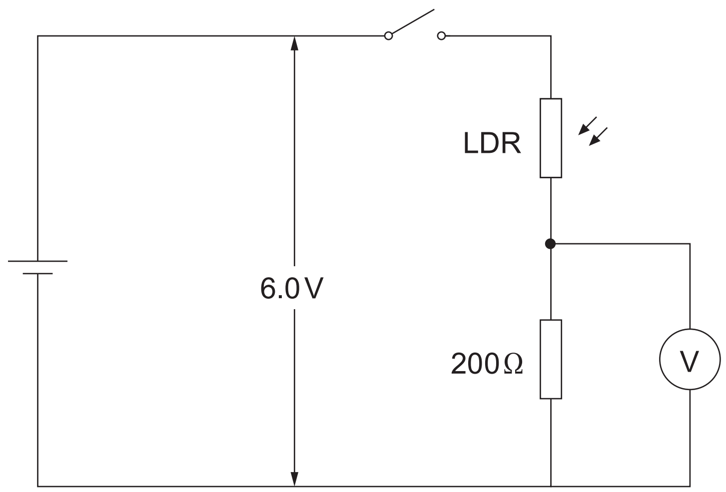
(c) The circuit shown includes a light-dependent resistor (LDR), whose resistance decreases as the intensity of light falling on it increases. A 6.0 V supply of negligible internal resistance is used.



(i) Calculate the voltmeter reading when the resistance of the LDR is 850 Ω and the switch is closed. [2]

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(ii) Explain, in clear steps, whether the voltmeter reading will increase or decrease when the intensity of light is increased. [2]

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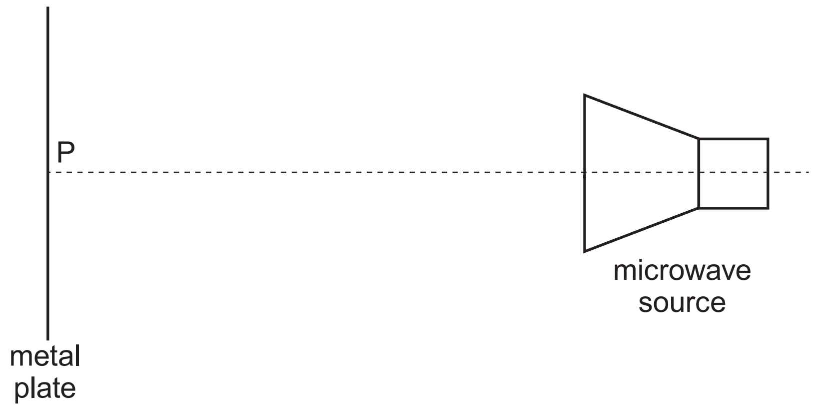
Examiner only

6. *(a)* In the set-up shown a series of *antinodes* is detected, at the following distances from the metal plate: 8 mm, 24 mm, 40 mm, 56 mm, 72 mm.



NOT TO SCALE

(i) Referring to the diagram, explain in terms of *interference* how an antinode is produced. [2]

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(ii) From the data, determine whether there is a node or an antinode at point **P** (on the metal plate). Give your reasoning in terms of wavelength. [2]

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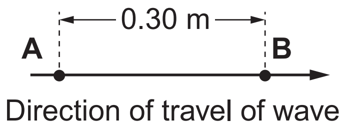
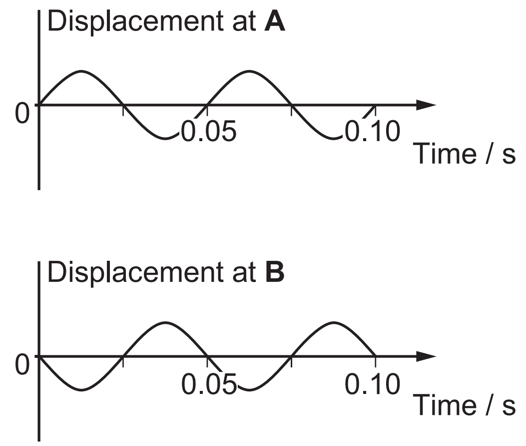
Examiner only

(b) Displacement–time graphs are given for two points, **A** and **B**, 0.30 m apart, in the path of a **progressive** wave.





(i) State the phase relationship between the displacements at **A** and **B**, then determine the longest, and the second longest, wavelength that the wave could have. [3]

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(ii) The speed of the waves is known to be between 10 m s^{-1} and 15 m s^{-1}. Determine which of the two wavelengths in *(b)*(i) is the correct one, giving your reasoning. [3]

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Examiner only

7. *(a)* (i) Draw the circuit diagram for an investigation of how the current through a filament lamp varies with the applied potential difference. [2]

(ii) The lamp is labelled "3 V, 0.16 A". The ammeter to be used is a multimeter with a 0 – 200 mA range and a 0 – 10 A range. State which range should be selected, and justify your choice. [1]

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(iii) The lamp has already been investigated by a student, Sion, who plotted the graph reproduced below. State **two** ways in which his investigation (**not** his graph plotting) could have been improved. [2]

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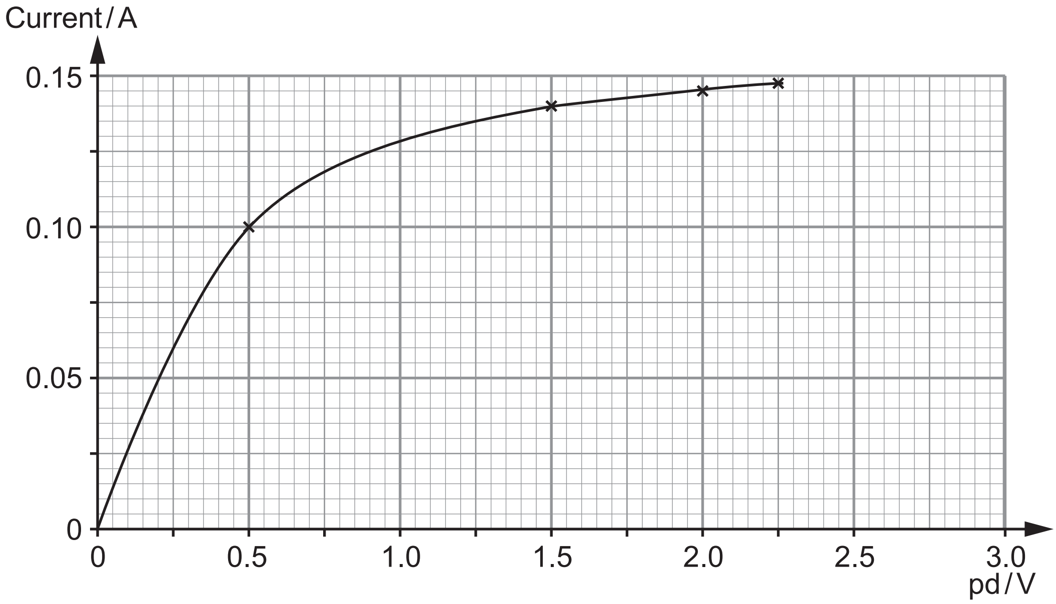

Examiner only

(b) (i) Use the graph to calculate the ratio: $\dfrac{\text{Resistance of lamp at } 2.00\,\text{V}}{\text{Resistance of lamp at } 0.50\,\text{V}}$ [3]

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(ii) Explain, in terms of free electrons, why the temperature **and** the resistance of a metal wire increase when the potential difference across the wire is increased. [6 QER]

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TURN OVER FOR THE LAST PART OF THE QUESTION

Examiner only

(c) A scientist claims to have made a material which is superconducting at room temperature. He publishes his procedure. Several research teams try to follow the same procedure, but without success. Discuss the arguments for and against spending more public money following up the scientist's claim. [3]

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END OF PAPER

Surname	Centre Number	Candidate Number
Other Names		2

GCE AS/A LEVEL – NEW

2420U20-1

S17-2420U20-1

PHYSICS – AS unit 2
Electricity and Light

THURSDAY, 8 JUNE 2017 – AFTERNOON

1 hour 30 minutes

For Examiner's use only		
Question	**Maximum Mark**	**Mark Awarded**
1.	**6**	
2.	**8**	
3.	**8**	
4.	**11**	
5.	**12**	
6.	**10**	
7.	**12**	
8.	**13**	
Total	**80**	

2420U201
01

ADDITIONAL MATERIALS

In addition to this examination paper, you will require a calculator and a **Data Booklet**.

INSTRUCTIONS TO CANDIDATES

Use black ink or black ball-point pen. Do not use pencil or gel pen. Do not use correction fluid.

Write your name, centre number and candidate number in the spaces at the top of this page.

Answer **all** questions.

Write your answers in the spaces provided in this booklet. If you run out of space use the continuation page(s) at the back of the booklet taking care to number the question(s) correctly.

INFORMATION FOR CANDIDATES

The total number of marks available for this paper is 80.

The number of marks is given in brackets at the end of each question or part-question.

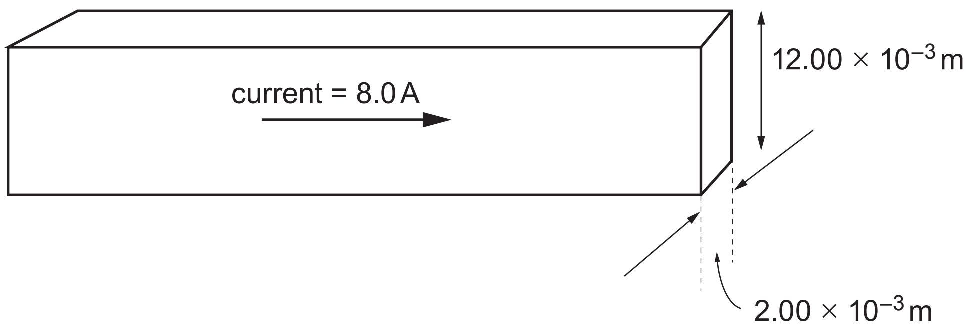
The assessment of the quality of extended response (QER) will take place in question **5**(*a*).

JUN172420U20101

Examiner only

Answer **all** *questions.*

1. A strip of metal of rectangular cross-section is shown.



(a) Calculate the drift velocity of free electrons in the strip when there is a current of 8.0 A as shown. The free electron concentration, n, in the metal is $8.5 \times 10^{28}\,\mathrm{m^{-3}}$. [3]

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(b) The resistance of the strip is 0.15 Ω from end to end. Calculate the energy transferred from electrical potential to thermal in the strip in a time of 20 s, for a current of 8.0 A, **and** briefly state how the transfer takes place. [3]

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2420U201 03

Examiner only

2. A student is given wire cutters and a reel of metal wire, and asked to find the resistivity of the metal. She cuts off a piece of the wire and makes measurements on it. She takes repeat readings and obtains the same values each time.

Quantity	Measurement	Instrument	Resolution of instrument
length	812 mm	metre ruler	1 mm
diameter	0.48 mm	digital calipers	0.01 mm
resistance	2.2 Ω	digital multimeter	0.1 Ω

(a) (i) Calculate the resistivity of the metal of the wire. [3]

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(ii) Calculate the **absolute** uncertainty in the resistivity giving your value to an appropriate number of significant figures. [3]

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Examiner only

(b) Suggest **one** way in which the student could reduce the uncertainty in her value for the resistivity, using the same reel of wire and the same instruments as before. Explain briefly why the uncertainty would be reduced. [2]

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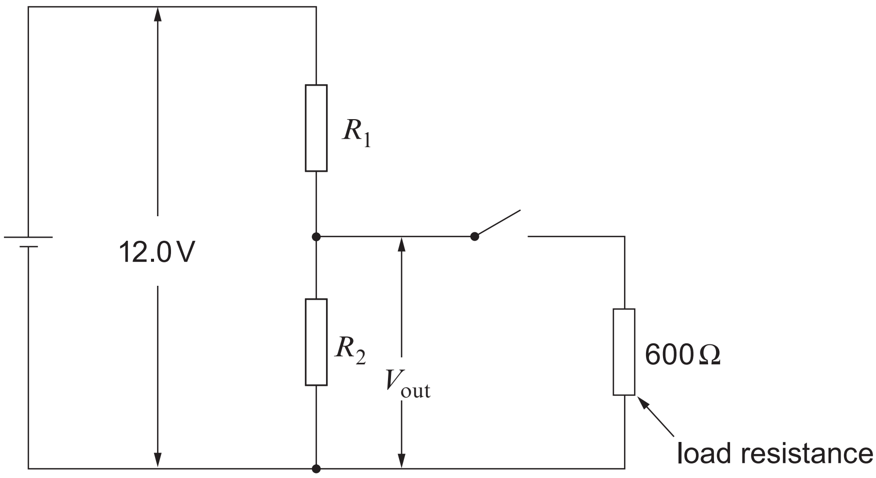
2420U201 05

Examiner only

3. For the potential divider circuit shown a student is asked to select values for R_1 and R_2 so that the circuit meets certain requirements.



Requirements

A an output pd, V_{out}, of 3.0 V when unloaded (switch open),
B a power dissipation of less than 0.50 W in R_1 when unloaded (switch open),
C a **decrease** of less than 0.40 V in V_{out} when the switch is closed.

Showing your working clearly, determine whether or not the circuit meets the requirements when R_1 = 450 Ω and R_2 = 150 Ω.

(a) Requirement **A**: an output pd, V_{out}, of 3.0 V when unloaded (switch open). [2]

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(b) Requirement **B**: a power dissipation of less than 0.50 W in R_1 when unloaded (switch open). [2]

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Examiner only

(c) Requirement **C**: a **decrease** of less than 0.40 V in V_{out} when the switch is closed. [4]

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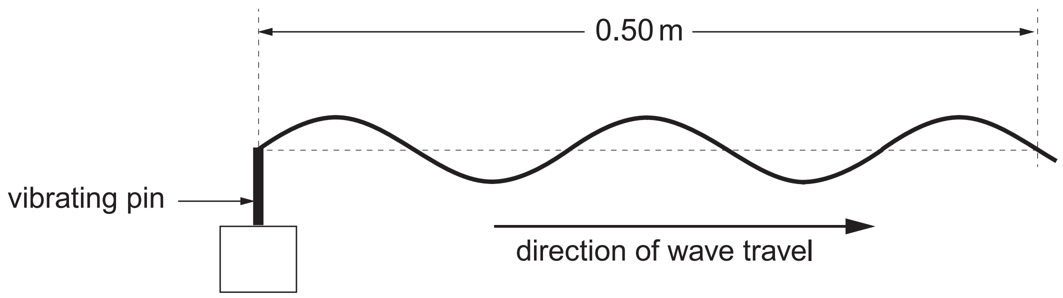
2420U201
07

Examiner only

4. *(a)* A **progressive** wave generated by a vibrating pin is travelling on a long string. The periodic time is 0.040 s. The diagram shows a part of the string at time t = 0.



(i) Calculate the distance that the wave travels in a time of 0.34 s. [3]

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(ii) Carefully sketch **on the diagram above** the same part of the string at time t = 0.34 s. (The periodic time is 0.040 s.) [1]

(b) If the string above is clamped rigidly a little way to the right of the part shown, a **stationary** wave may be observed.

(i) State what is meant by a **node** in a stationary wave, and state how far apart the nodes will be in *this* stationary wave. [2]

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(ii) State how, if at all, the *phase* of the oscillations of the particles of the string varies with distance along the string, for a stationary wave. [2]

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Examiner only

(iii) Explain, in terms of *interference*, how the stationary wave is produced for this stationary wave on the string. [2]

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(iv) Stationary waves may also be observed on the string when the wave source is set to higher frequencies. State **one** way, apart from having a different frequency, in which the stationary waves will be different from the original stationary waves. [1]

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2420U201 09

Examiner only

5. *(a)* Describe in detail how you would determine the wavelength of light from a (low-power) laser, given an opaque plate with two parallel slits. (The distance between the centres of the slits is less than 1 mm and has already been measured.) [6 QER]

Examiner only

(b) A diffraction grating is used with a laser as shown, and a line of bright spots is seen on the screen as shown.

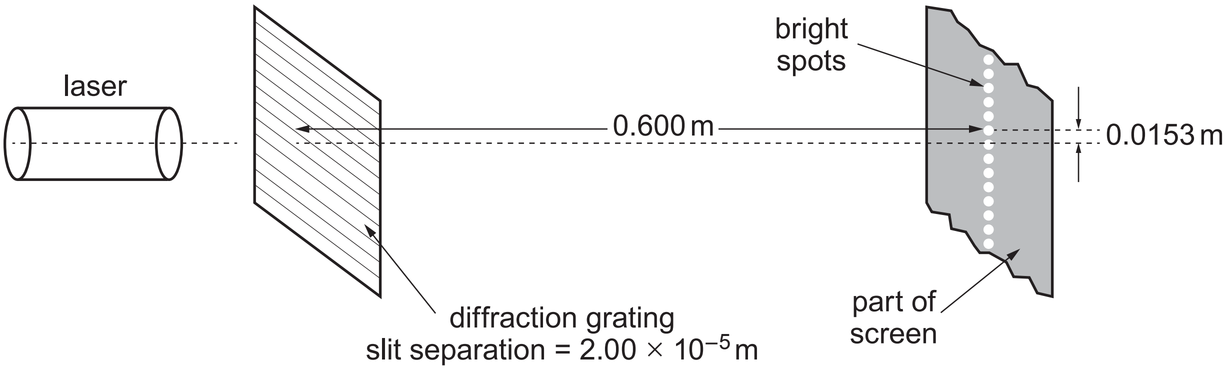

DIAGRAM NOT TO SCALE

(i) Using a labelled sketch of a relevant triangle (in the space below), explain why, to a very good approximation:

$$\sin\theta_1 = \frac{0.0153}{0.600}$$

in which θ_1 is the angle between the zeroth order and first order beams. [2]

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(ii) Determine the wavelength of light from the laser. [2]

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Examiner only

(iii) Calculate the highest **order** produced by this grating for this wavelength, and suggest why so many orders are produced by this grating. [2]

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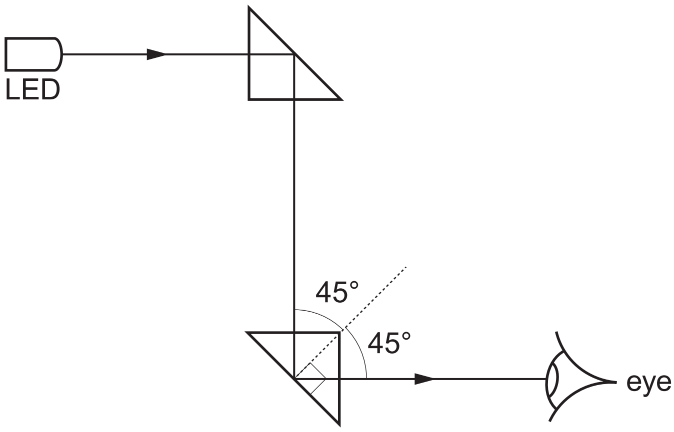
Examiner only

6. *(a)* A student sets up a simple periscope consisting of two glass prisms (n_{glass} = 1.52). She tests it by using it to view a light emitting diode (LED) as shown.



(i) Give the full name of the process by which the light changes direction in the prisms. [1]

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(ii) The student wonders whether the periscope will still work properly if the lower prism is surrounded by water (n_{water} = 1.33). Determine whether or not it will, giving your reasoning clearly. [3]

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(b) (i) A particular monomode fibre has a core of refractive index 1.515. Calculate the time it takes for a pulse of light to travel through 1.50 km of the fibre. [3]

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Examiner only

(ii) Explain the advantage of a monomode fibre over a fibre with a much thicker core, for the transmission of a rapid stream of data. [3]

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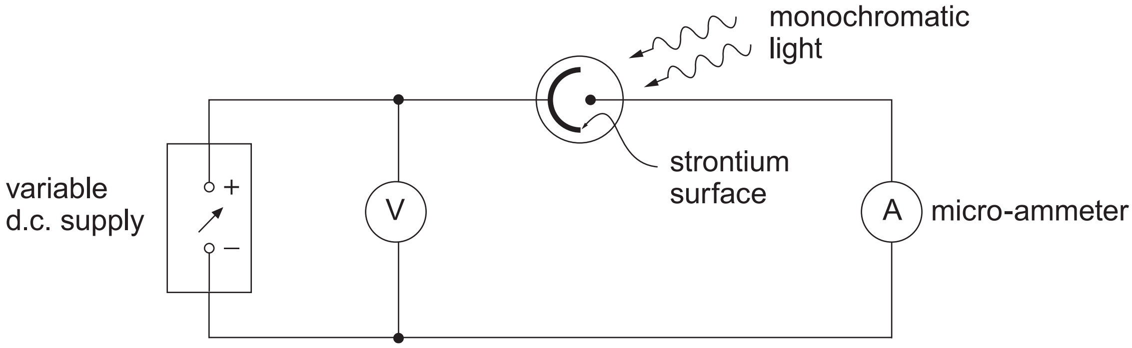
Examiner only

7. *(a)* A student uses the apparatus shown in the diagram to determine the maximum kinetic energy, $E_{\mathrm{k\,max}}$, of electrons ejected from a strontium surface by light of wavelength 405 nm.



He records that $E_{\mathrm{k\,max}}$ = 0.480 eV.

(i) Describe briefly how he used the apparatus to obtain this result. [2]

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(ii) Determine a value for the work function of strontium from this result. [3]

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Examiner only

(iii) The student next attempts to measure $E_{k\,max}$ for the same surface, but with light of wavelength 650 nm. He is surprised to find that the micro-ammeter reads zero for **all** applied pds – even when he makes the light brighter. Explain in terms of photons why he should not be surprised, justifying your answer with a calculation. [3]

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(b) (i) Calculate the speed at which an electron must be moving to have a de Broglie wavelength of 5.0×10^{-11} m. [2]

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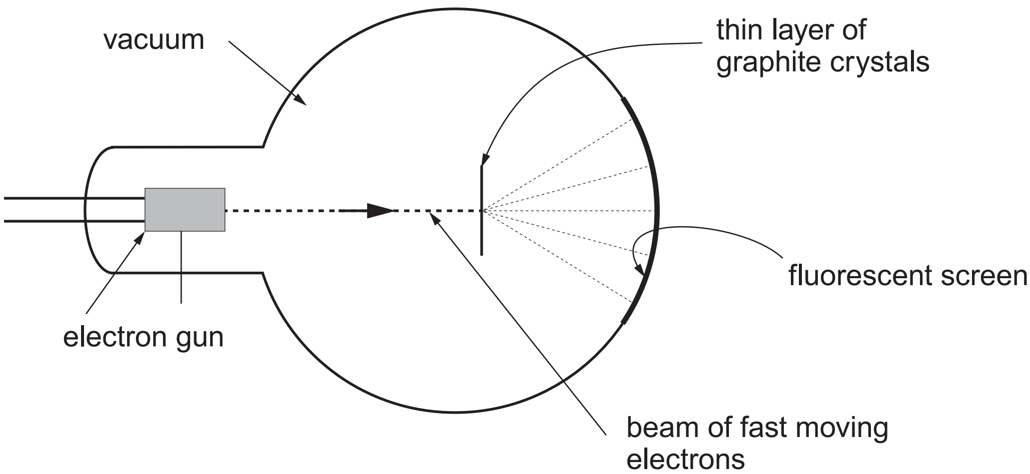
Examiner only

(ii) The diagram shows one way in which an aspect of the nature of electrons may be demonstrated by experiment.



Several bright concentric circles, caused by the impact of electrons on the fluorescent screen are seen. Explain briefly how the pattern of circles arises. [2]

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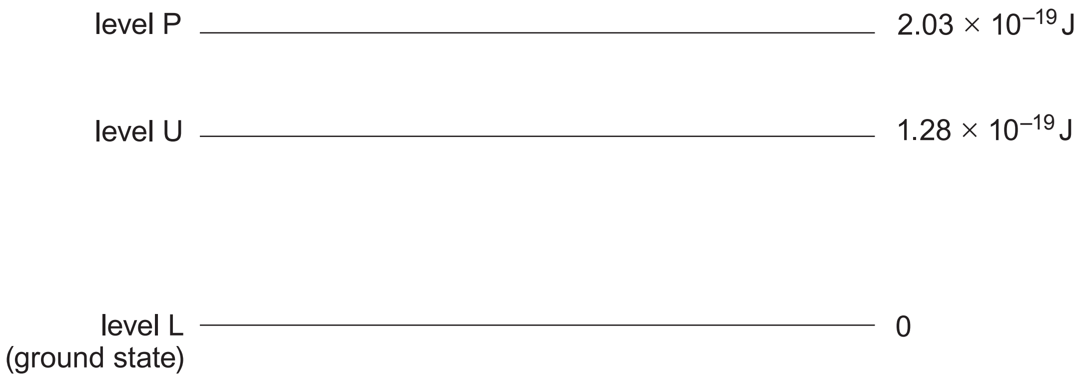
Examiner only

8. *(a)* A simplified energy level diagram for a 3-level laser system is given.



(i) Calculate the wavelength of photons which can take part in transitions between levels U and L. [2]

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(ii) Describe the processes of *absorption* and *stimulated emission* involving levels U and L. [4]

absorption: ....

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stimulated emission: ....

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(iii) If the relative populations of levels P, U and L are 0.2%, 54.0% and 45.8%, state with a reason whether absorption or stimulated emission is the more likely event for a photon of the wavelength calculated in *(a)*(i). [1]

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Examiner only

(iv) The laser is pumped by promoting electrons from the ground state to level P. Explain why electrons must spend only a very short time in level P, but a relatively long time in level U. [2]

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(b) It has been suggested that energy should be carried across the country as high power light beams from lasers. Discuss whether or not this might be a good alternative to using electrical power lines. [4]

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END OF PAPER

Surname	Centre Number	Candidate Number
Other Names		2

AS/A LEVEL

2420U20-1

S18-2420U20-1

PHYSICS – AS unit 2
Electricity and Light

FRIDAY, 18 MAY 2018 – MORNING

1 hour 30 minutes

For Examiner's use only		
Question	**Maximum Mark**	**Mark Awarded**
1.	**15**	
2.	**7**	
3.	**10**	
4.	**8**	
5.	**7**	
6.	**8**	
7.	**12**	
8.	**13**	
Total	**80**	

2420U201
01

ADDITIONAL MATERIALS

In addition to this paper you will require a calculator and a **Data Booklet**.

INSTRUCTIONS TO CANDIDATES

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INFORMATION FOR CANDIDATES

The total number of marks available for this paper is 80.
The number of marks is given in brackets at the end of each question or part-question.
The assessment of the quality of extended response (QER) will take place in question **7*(b)***.

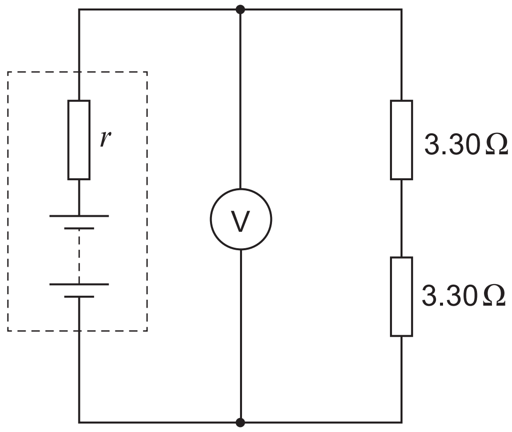
MAY182420U20101

Examiner only

Answer **all** *questions.*

1. *(a)* Two 3.30 Ω resistors are connected in series across a battery of emf 4.80 V as shown. The voltmeter reads 4.33 V.

Diagram 1



(i) State in terms of work or energy what is meant by the emf of a battery. [2]

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(ii) Show that the internal resistance, r, of the battery is approximately 0.7 Ω. [2]

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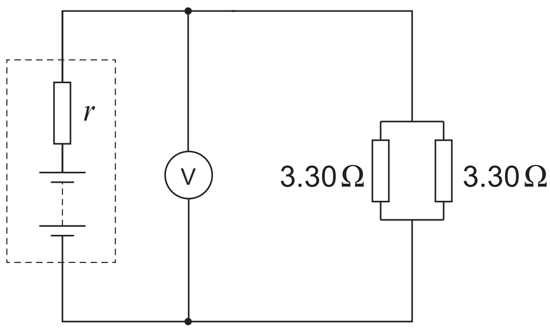
(iii) When the resistors are connected in parallel as shown below, the voltmeter reads 3.35 V.

Diagram 2



 (2420U20-1)

Examiner only

(I) **Without further calculation** explain why you would expect the reading to be lower, now that there is a lower resistance across the cell terminals. [2]

(II) Calculate the number of electrons entering **either** of the resistors (shown in Diagram 2) per **minute**. [3]

2420U201 03

(b) The heating element (a coil of wire) in an electric heater dissipates energy at a rate of 1.00 kW, when connected across the 230 V mains supply. Calculate:

(i) the resistance of the coil; [2]

(ii) the energy dissipated per hour, giving your answer in megajoules (MJ). [1]

Examiner only

(c) For each megajoule of heat from an electric heater, approximately 0.08 m^3 of gas would have to be burned in a gas-fired electricity power station. For each megajoule of heat from a domestic gas fire or boiler, approximately 0.03 m^3 of gas is burned. Discuss whether the use of electric heaters in houses should be discouraged. Calculations are not required. [3]

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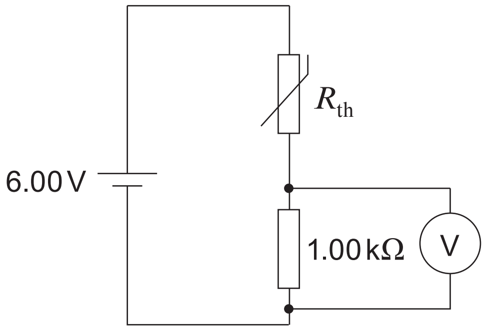
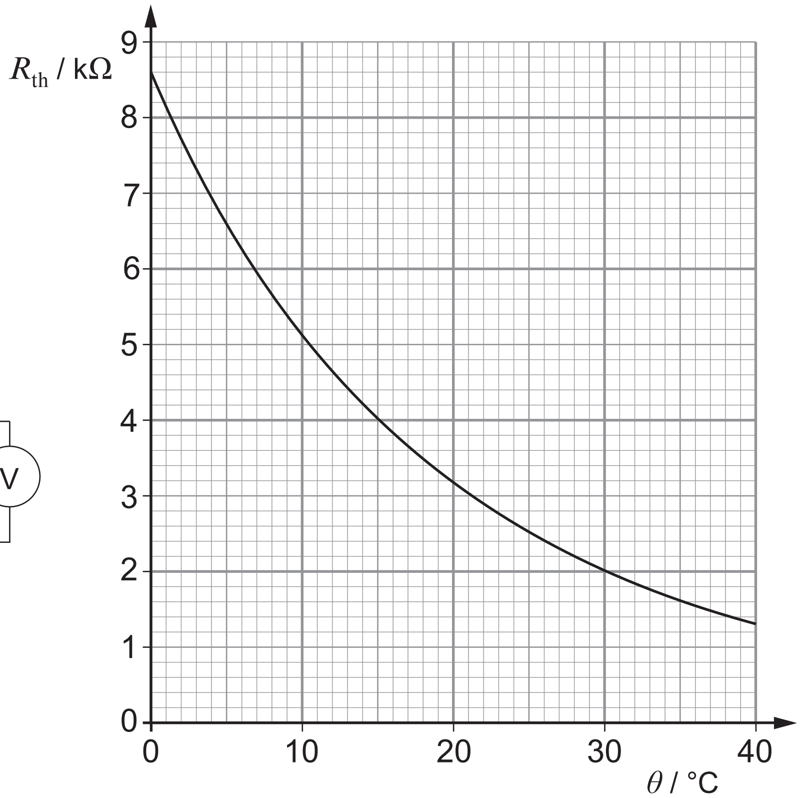
Examiner only

2. The resistance, R_{th}, of the thermistor in the circuit varies with temperature as shown in the graph.





(a) Determine the temperature of the thermistor when the voltmeter reads 1.20 V. [3]

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Examiner only

(b) The set-up can be used as a thermometer, with the thermistor used as a temperature probe. David suggests that there must be a simple rule of the type "an increase of 0.10 V in the voltmeter reading corresponds to a temperature increase of n °C in which n is a constant." Without further calculation, discuss whether he is right. [2]

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(c) Explain why the current through the thermistor must be very low, in order for it to work properly as a probe to measure the temperature of its surroundings. [2]

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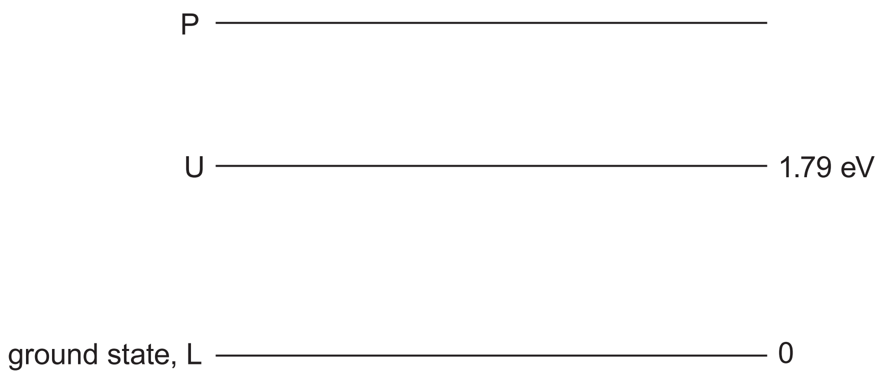
Examiner only

3. A simplified energy level diagram for a 3-level laser system is given.



(a) (i) Calculate the wavelength of light emitted by stimulated emission. [3]

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(ii) Assuming that some photons of this wavelength are already present in the laser cavity, explain why a population inversion is needed for light amplification to take place. [2]

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Examiner only

(b) The laser produces a light beam of power 6.0 mW.

(i) Show that the number of photons emitted per second is approximately $2 \times 10^{16}\ s^{-1}$. [2]

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(ii) Calculate the momentum of the light leaving the laser per second. [2]

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(iii) Calculate the force exerted by the light beam on a perfectly reflecting surface, if it strikes the surface normally. [1]

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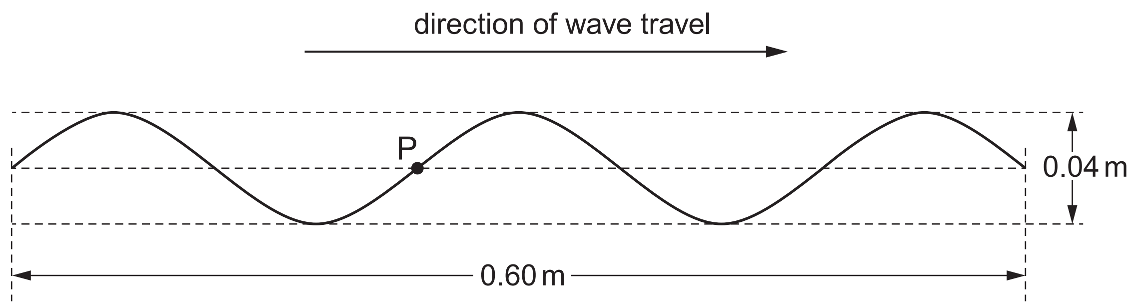
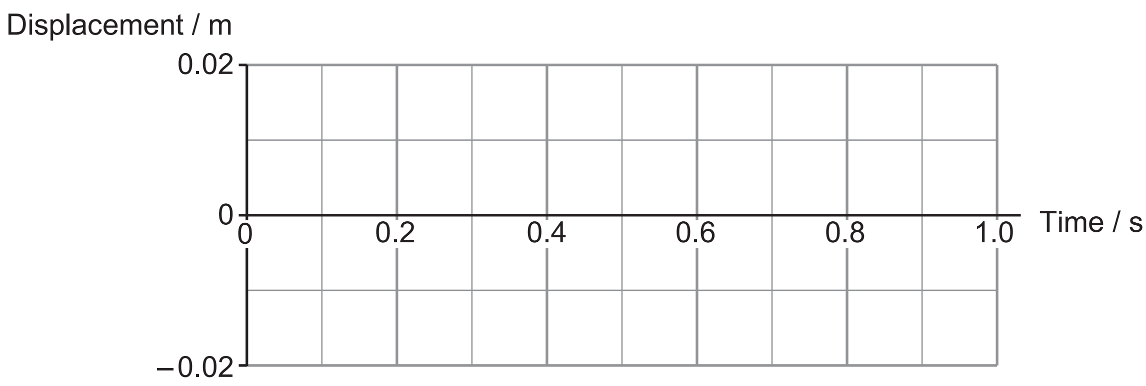
Examiner only

4. *(a)* A progressive wave is travelling from left to right along a stretched string at a speed of 0.40 m s^{-1}. The diagram shows the string at time t = 0.



Carefully sketch, **on the grid below**, a displacement–time graph for point **P** on the string between t = 0 and t = 1 s. The space below the grid is for your working. [3]

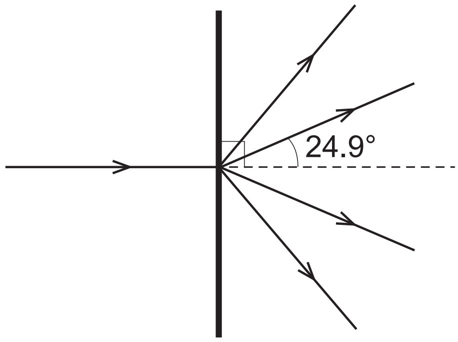


Space for working:

 (2420U20-1)

Examiner only

(b) The distance between the centres of the slits in a diffraction grating is 1500 nm. Monochromatic light is shone normally on to the grating.



(i) First order beams emerge at angles of 24.9° to the normal (see diagram). Calculate the wavelength of the light. [2]

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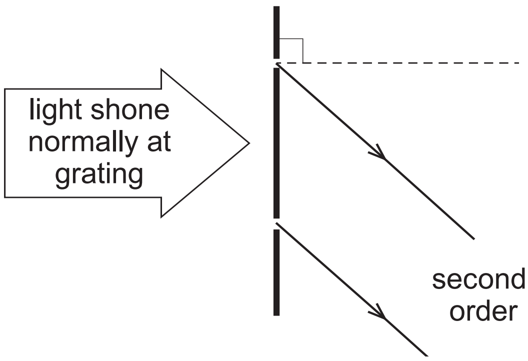
(ii) Explain **in terms of path difference** why the **second** order beams emerge from the diffraction grating at 57.4° to the normal. You will need to add to the diagram (which shows two adjacent slits in the grating). [3]



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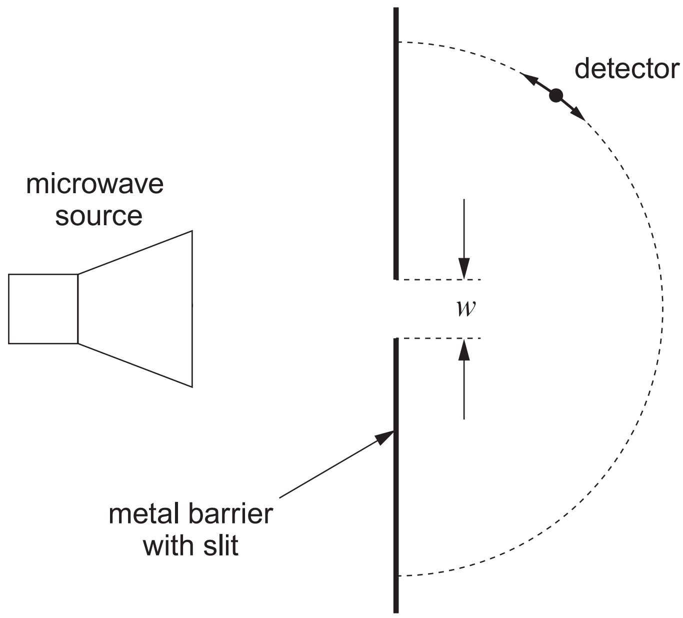
Examiner only

5. *(a)* In the set-up shown, a microwave detector moved along the semi-circular path detects diffracted waves at **all** points around the semicircle.



(i) The width, w, of the gap is 25 mm. What can be deduced about the wavelength of the microwaves? [1]

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(ii) Rhian suggests that making the gap wider will increase the intensity of the microwaves detected. Discuss to what extent she is correct. [2]

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Examiner only

(b) In the set-up shown, slits S_1 and S_2 act as in-phase sources. Rhian detects **minima** of wave intensity at P and Q, and a single maximum between P and Q.

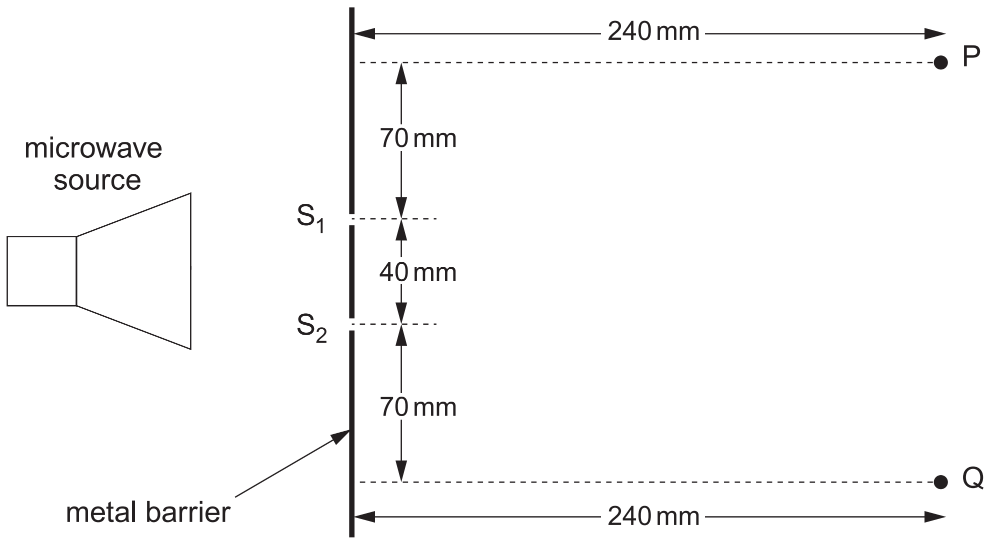

Diagram not to scale

Calculate the path lengths S_1P and S_2P, and **hence** the wavelength of the microwaves. [4]

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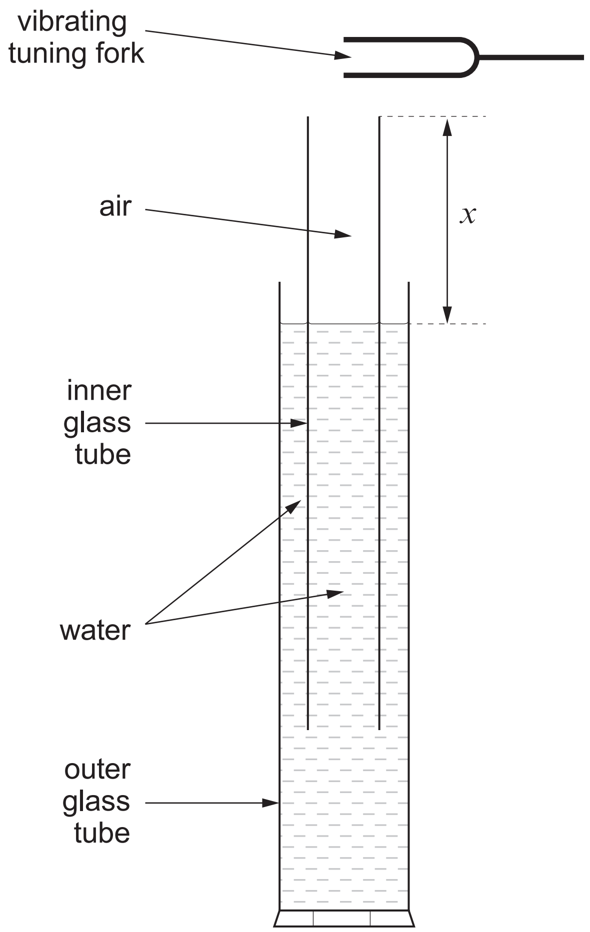
6. Matthew uses the apparatus shown to find a value for the speed of sound in air using stationary waves.



Starting with a very low value of distance x, he gradually raises the inner glass tube. At a certain value of x, the note of the tuning fork is heard loudly. Matthew then measures and records x.

He repeats the procedure several times, obtaining these results.

x / m	0.327	0.329	0.322	0.328	0.332	0.323

The tuning fork frequency, f, is given as 256.0 Hz ± 1%.

Examiner only

(a) The speed of sound in air, v_s, can be found from the equation:

$$v_s = 4fx$$

(i) Calculate the mean value of x along with its **percentage** uncertainty. [2]

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(ii) Hence calculate the value of v_s along with its **absolute** uncertainty. [2]

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Examiner only

(b) The loud sound is due to stationary waves of frequency, f, in the air column. There is a node of displacement at the water surface and an antinode of displacement close to the open end (top) of the inner glass tube.

(i) Derive the equation: [2]

$$v_s = 4fx$$

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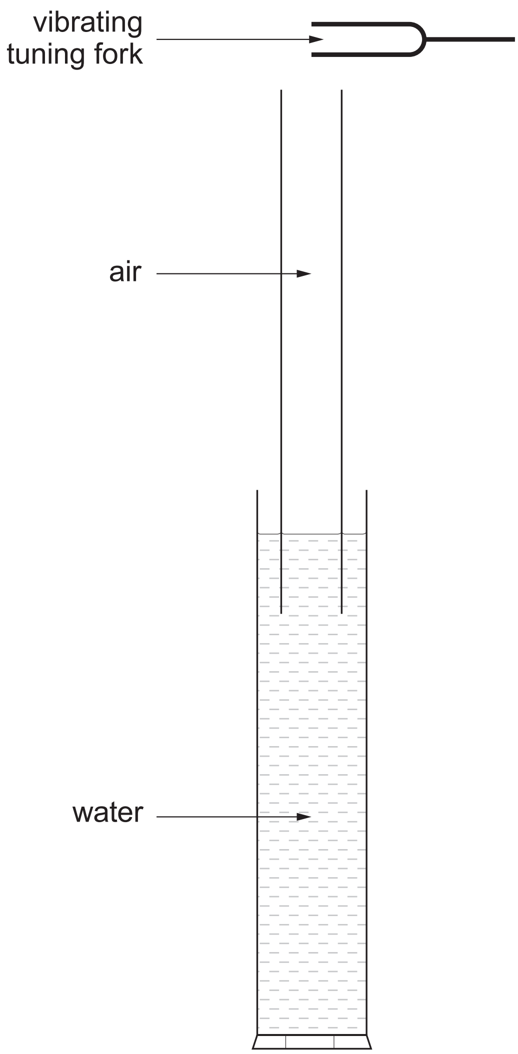
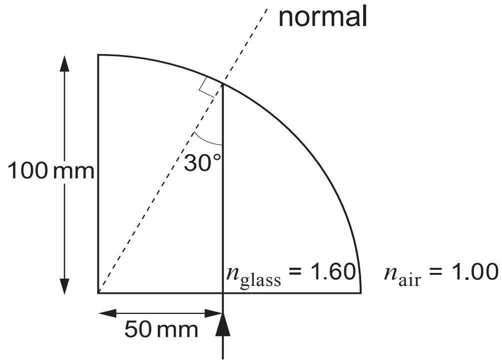
(ii) If the inner glass tube is raised considerably further a second distance is found at which a loud note of frequency, f, is heard. This again corresponds to a stationary wave in the air column. Show the positions of its nodes and antinodes of displacement **on the diagram below**. [2]



8

Examiner only

7. *(a)* (i) A narrow beam of light enters a glass block of quarter-circle cross-section, as shown. The refractive index of the glass is 1.60.



Calculate the angle of refraction into the air at the curved surface and carefully sketch the refracted beam **on the diagram**. [Use of a protractor is not required.][3]

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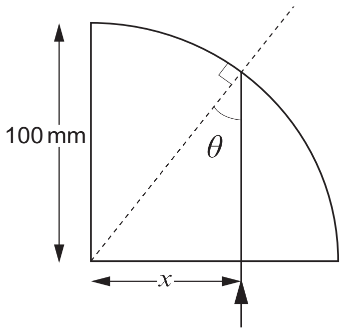
(ii) Calculate the largest distance, x, along the bottom face of the block, at which the beam can enter the block normally, for it to emerge from the curved face. You should refer to the angle θ in your calculation. [3]



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Examiner only

(b) Explain how a multimode (thick) glass fibre transmits light, and why rapid streams of data cannot be transmitted successfully through long lengths of multimode fibre. [6 QER]

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Examiner only

8. *(a)* (i) Light of frequency less than $\frac{\phi}{h}$ cannot eject electrons from a surface of work function ϕ, even if the light intensity is increased. Explain this in terms of photons. [3]

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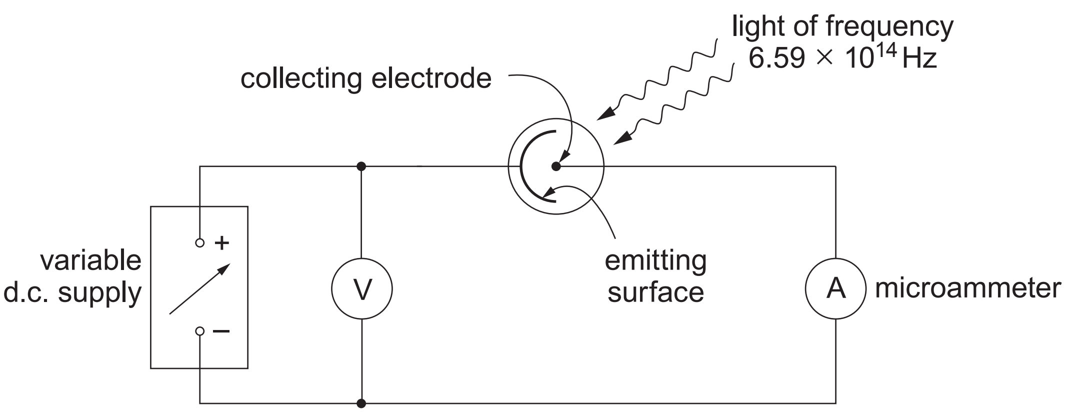
(ii) The emitting surface in a vacuum photocell is known to be made of one of the metals listed below (with their work functions).

Metal	caesium	potassium	barium	calcium	zinc
Work function / 10^{-19} J	3.12	3.68	4.03	4.59	5.81

The photocell is included in the circuit shown, and illuminated with light of frequency 6.59×10^{14} Hz.



With zero pd applied, the microammeter indicates a current. At some pd between 0 V and 0.35 V the microammeter reading drops to zero.

Examiner only

Determine which metal the emitting surface is made of, giving your reasoning clearly. [4]

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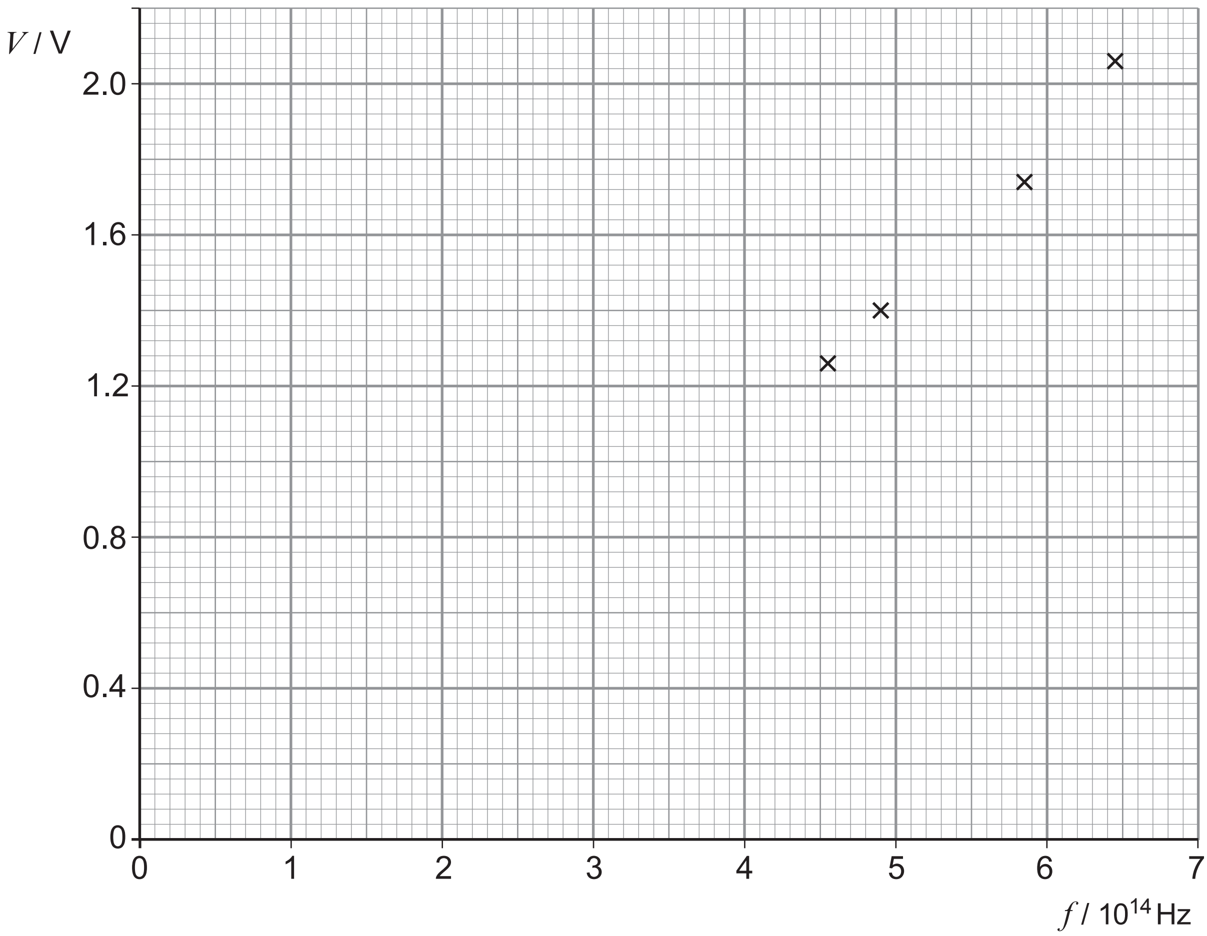
TURN OVER FOR THE REST OF THE QUESTION

Examiner only

(b) Rachel varies the pd across a light-emitting diode (LED) and notes the value, V, for which she can just see light from the LED. She also notes the frequency, f, of the light, as supplied by the LED's makers. She does the same for three other LEDs and plots V against f (below).



It has been suggested that V and f are related by the equation:

$$V = \frac{h}{e} f$$

(i) **On the graph** draw the line of best fit. [1]

(ii) Discuss the extent to which the graph supports an equation of this form. [2]

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Examiner only

(iii) Determine the gradient of the graph, and hence a value for h to an appropriate number of significant figures. Assume that the equation predicts $\frac{\Delta V}{\Delta f}$ correctly. [3] Show your working clearly.

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END OF PAPER

Surname	Centre Number	Candidate Number
Other Names		2

AS/A LEVEL

2420U20-1

S19-2420U20-1

PHYSICS – AS unit 2
Electricity and Light

FRIDAY, 17 MAY 2019 – MORNING

1 hour 30 minutes

For Examiner's use only		
Question	**Maximum Mark**	**Mark Awarded**
1.	**10**	
2.	**10**	
3.	**9**	
4.	**19**	
5.	**9**	
6.	**14**	
7.	**9**	
Total	**80**	

2420U201
01

ADDITIONAL MATERIALS

In addition to this paper you will require a calculator, ruler and a **Data Booklet**.

INSTRUCTIONS TO CANDIDATES

Use black ink or black ball-point pen. Do not use pencil or gel pen. Do not use correction fluid.

Write your name, centre number and candidate number in the spaces at the top of this page.

Answer **all** questions.

Write your answers in the spaces provided in this booklet. If you run out of space use the additional page at the back of the booklet taking care to number the question(s) correctly.

INFORMATION FOR CANDIDATES

The total number of marks available for this paper is 80.

The number of marks is given in brackets at the end of each question or part-question.

The assessment of the quality of extended response (QER) will take place in question **4*(a)***.

Examiner only

Answer **all** *questions.*

1. *(a)* (i) State Ohm's law. [1]

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(ii) What can be said about the *resistance* of a conductor that obeys Ohm's law? [1]

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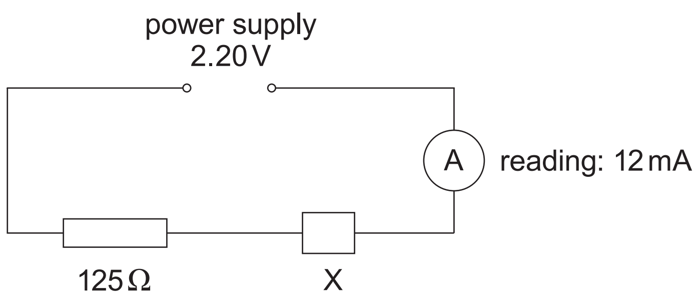
(b) (i) An electrical component, X, is included in the circuit shown. The internal resistance of the power supply is negligible.



Show that the resistance of X in this circuit is approximately 60 Ω. [2]

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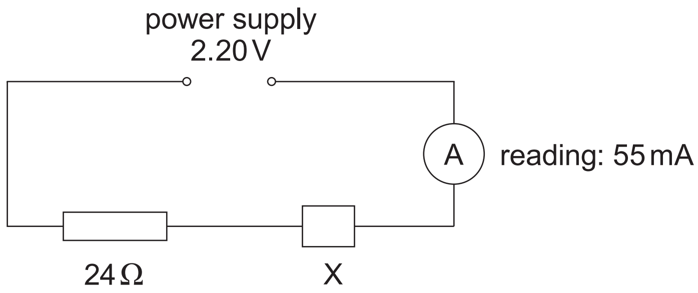
Examiner only

(ii) When the 125 Ω resistor in *(b)*(i) is replaced by a 24 Ω resistor, the reading on the ammeter increases, as shown below.



Evaluate whether or not X obeys Ohm's law, presenting your argument clearly. [2]

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(iii) State, giving a reason, whether or not X could be a filament lamp. [1]

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(c) A certain high temperature superconductor has a transition temperature of –188 °C. The boiling point of liquid nitrogen is –196 °C.

(i) State what is meant by the *transition temperature* of a superconductor. [1]

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(ii) Give one possible use for a high temperature superconductor and state why it would be an advantage for the transition temperature to be above the boiling point of liquid nitrogen. [2]

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2420U201 03

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Examiner only

2. *(a)* A battery does not *store* charge. State what a battery *does* do in relation to charge in an electric circuit. [1]

(b) A battery consists of three cells, **each** of emf 1.60 V and internal resistance 0.10 Ω, connected in series. The battery is connected to an electromagnet of resistance 1.20 Ω.

(i) Show clearly that the current is approximately 3 A. [The space is for a diagram if required.] [3]

(ii) Calculate the rate (in watts) at which:

I. energy is dissipated by the electromagnet; [1]

II. the battery's chemical energy is being used. [1]

(iii) The answer to *(b)*(ii)II. is expected to be greater than the answer to *(b)*(ii)I. Explain where the missing energy goes. [2]

Examiner only

(c) A student wishes to maximise the current through the electromagnet. She has a spare cell of emf 1.50 V and internal resistance 0.50 Ω. Evaluate whether or not she should put it in series with the battery, giving your calculations and conclusion clearly. [2]

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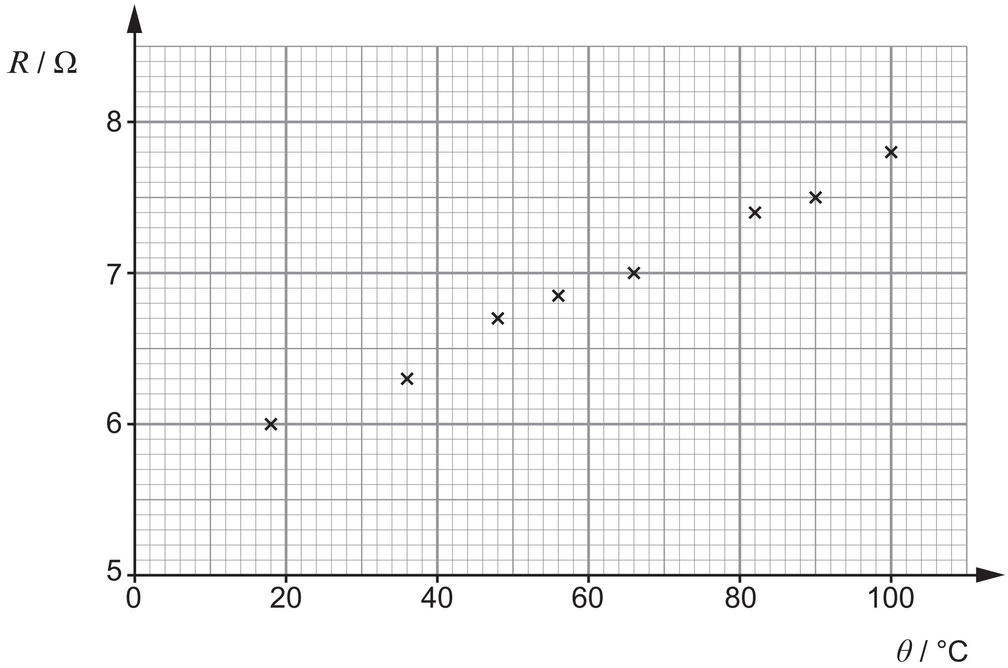
Examiner only

3. A student slowly heated a coil of insulated copper wire. He took readings of its temperature, θ, and resistance, R, at intervals. The readings are plotted below.



(a) Bearing in mind the range of temperatures, suggest how the student heated the coil, and how he could have extended the range of temperatures down to just above 0 °C. [2]

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(b) Textbooks state that the resistance, R, of a metal wire is related to its celsius temperature, θ, by an equation of the form:

$$R = R_0 \alpha\theta + R_0$$

in which R_0 and α are positive constants.

(i) Explain how the readings plotted support the equation. [3]

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Examiner only

(ii) Determine from the graph the values, with units, of:

I. R_0; [1]

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II. α. [3]

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2420U201 07

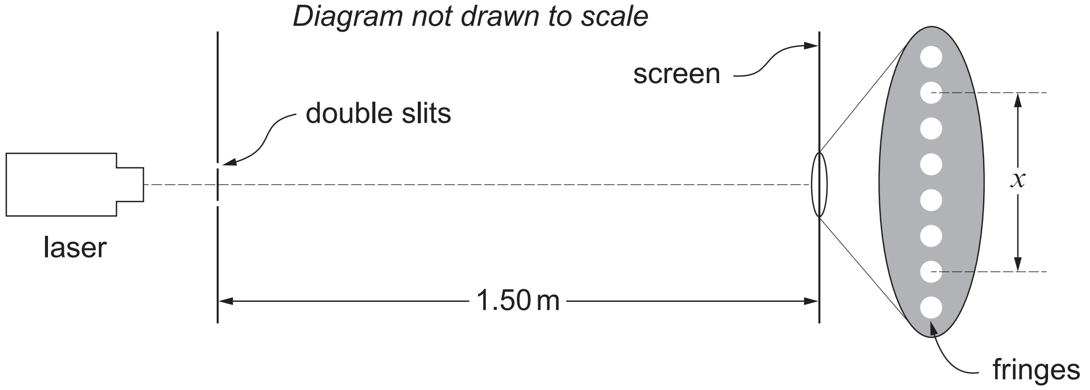
Turn over.

Examiner only

4. *(a)* Explain the conditions that are required of the light sources, for a two-source interference pattern to be observed. Include examples of when the requirements would and would not be met. [6 QER]

Examiner only

(b) A Young's fringes experiment is set up as shown.



(i) Six students measure the distance x, obtaining these results:

6.5 mm 6.3 mm 6.9 mm 6.9 mm 6.7 mm 6.4 mm

Calculate the mean value for fringe separation (the separation of the centres of **neighbouring** bright fringes), together with its **percentage** uncertainty. [4]

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(ii) The uncertainty in the distance of 1.50 m shown in the diagram is negligible. The separation between the centres of the slits is given by the makers of the double slit as 0.60 mm ±5 %. Calculate a value for the wavelength of the light from the laser, along with its **percentage** uncertainty. [3]

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QUESTION CONTINUES ON NEXT PAGE

Examiner only

(c) The laser beam is now shone normally on to a diffraction grating with centres of slits separated by 1 500 nm. Second order beams emerge at angles of 45° to the normal.

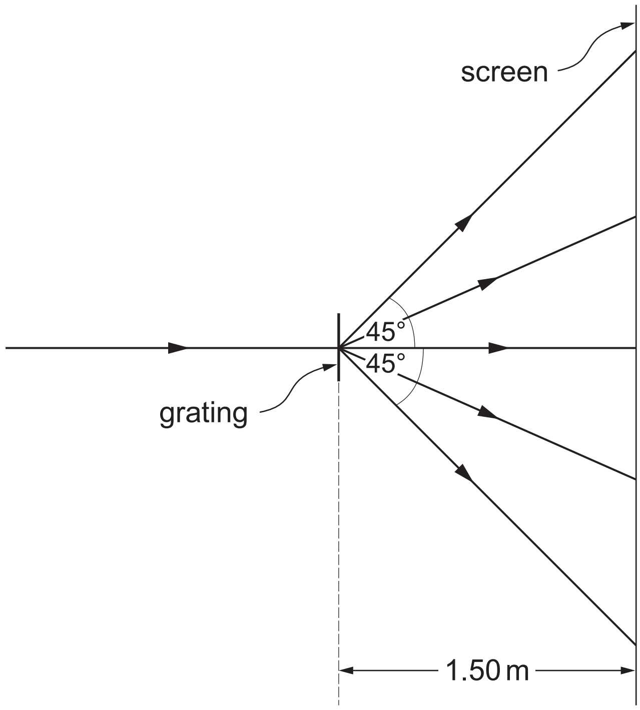

Diagram not drawn to scale

(i) Determine a value for the wavelength of the laser light from the data. [2]

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Examiner only

(ii) The beams from the grating strike the screen as shown opposite. The pattern on the screen may be compared with the pattern of fringes in part *(b)* (Young's experiment). There are fewer bright spots in the case of the grating. State two other differences between the patterns **and** state briefly why each difference occurs. [4]

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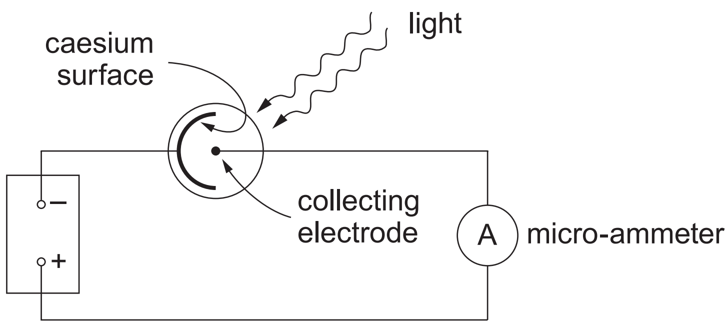
Examiner only

5. *(a)* Define the *work function* of a metal. [1]

(b) The work function of caesium is 3.2×10^{-19} J. Show that the frequency of light that will eject electrons from a caesium surface with a maximum kinetic energy of 1.5×10^{-19} J is approximately 7×10^{14} Hz. [2]

(c) The **same frequency** of light is shone on to the caesium surface in a vacuum photocell in the circuit shown.



The light energy falling each second on the caesium surface is 0.30 J.

(i) Show that the number of photons striking the caesium surface each second is approximately 6×10^{17}. [2]

Examiner only

(ii) The current indicated by the ammeter is 0.80 μA. Calculate the number of electrons per second emitted from the caesium surface, stating your assumption. [3]

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(iii) Hence calculate the *probability* of a photon of this frequency ejecting an electron from a caesium surface. [1]

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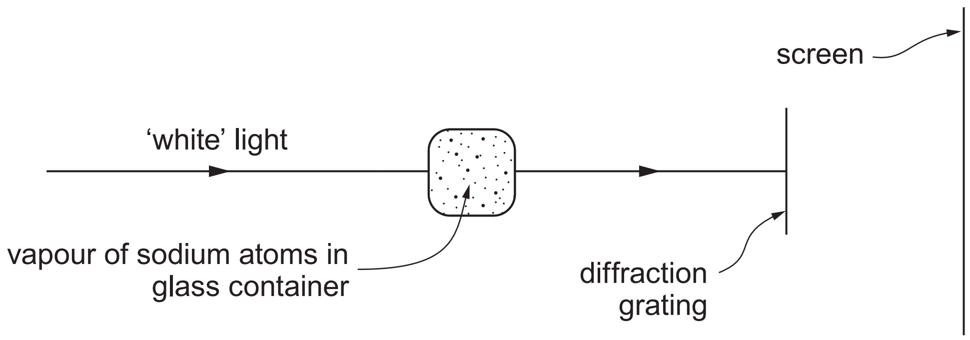
Examiner only

6. *(a)* The diagram shows how the *absorption spectrum* of sodium atoms may be produced.



Describe the appearance of the absorption spectrum, comparing it with the *emission spectrum* of sodium atoms. [3]

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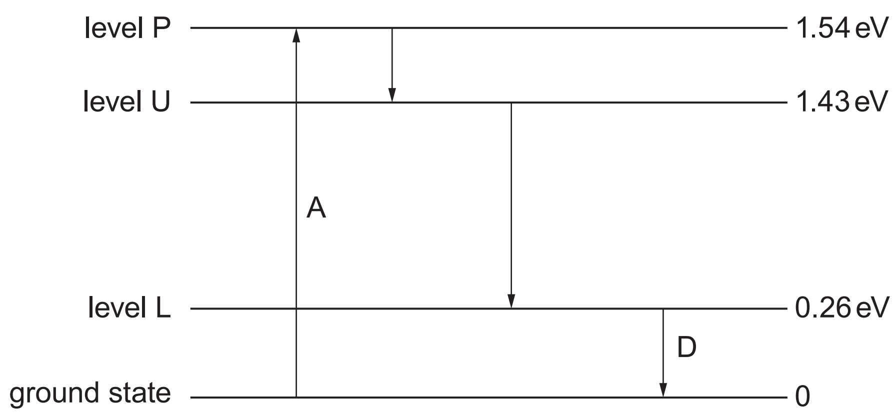
(b) A simplified energy level diagram for the amplifying medium of a 4 level laser is given.



Examiner only

(i) Referring to *populations*, explain the part played in the operation of the laser by:

I. transition A; [2]

II. transition D. [2]

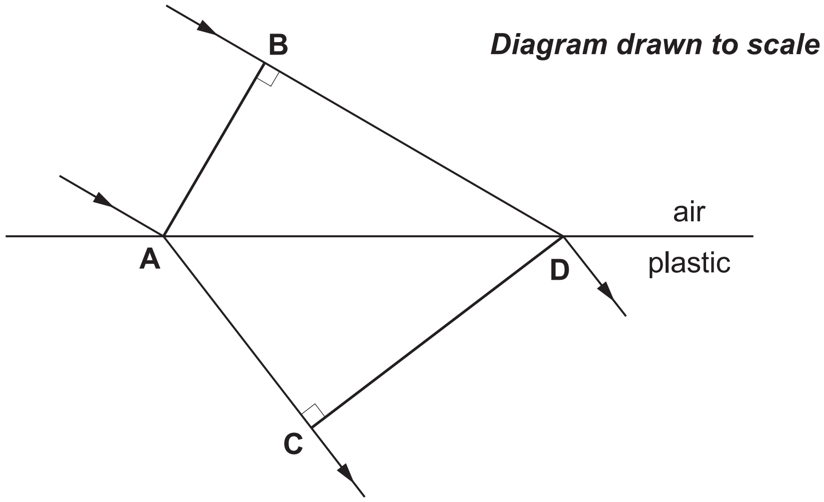
(ii) Calculate the wavelength of the radiation emitted by stimulated emission from the laser, and name the region of the electromagnetic spectrum in which it lies. [4]

(c) Most laser pointers produce polarised light. Discuss whether or not students attending a lecture in which a laser pointer is used should be given spectacles fitted with polarising filters (polaroids) to wear for safety during the lecture. [3]

14

Examiner only

7. *(a)* The diagram (**real size**) shows a beam of light passing from air into a clear plastic. AB is a wavefront about to enter the plastic, and CD is its position a short time later.



(i) Determine the speed of light in the plastic **using measurements made with a ruler** from the diagram, and taking the speed of light in air to be equal to its speed in a vacuum. Show your working clearly. [3]

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(ii) Calculate the refractive index of the plastic. [1]

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Examiner only

(b) In a multimode fibre of length 120.00 m, the length of the longest possible (zig-zag) route for light to travel successfully is 120.90 m. The refractive index of the fibre core is 1.520. Very short pulses of light are sent into one end of the fibre at intervals of 4.0 ns.

(i) Evaluate whether or not overlap (or overtaking) of pulses will occur before the pulses have reached the other end of the fibre. [3]

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(ii) Part of the longest possible successful zig-zag route is shown below.

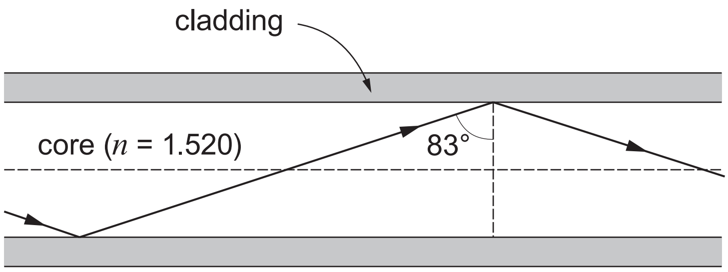

Diagram not drawn to scale

Calculate the refractive index of the cladding. [2]

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END OF PAPER

Unit 2 2022.pdf

Surname	Centre Number	Candidate Number
First name(s)		2

GCE AS/A LEVEL

2420U20-1

Z22-2420U20-1

MONDAY, 6 JUNE 2022 – MORNING

PHYSICS – AS unit 2
Electricity and Light

1 hour 30 minutes

For Examiner's use only		
Question	**Maximum Mark**	**Mark Awarded**
1.	**9**	
2.	**16**	
3.	**6**	
4.	**11**	
5.	**10**	
6.	**12**	
7.	**8**	
8.	**8**	
Total	**80**	

2420U201
01

ADDITIONAL MATERIALS

In addition to this paper you will require a calculator and a **Data Booklet**.

INSTRUCTIONS TO CANDIDATES

Use black ink or black ball-point pen. Do not use gel pen or correction fluid.
You may use a pencil for graphs and diagrams only.
Write your name, centre number and candidate number in the spaces at the top of this page.
Answer **all** questions.
Write your answers in the spaces provided in this booklet. If you run out of space use the additional page at the back of the booklet taking care to number the question(s) correctly.

INFORMATION FOR CANDIDATES

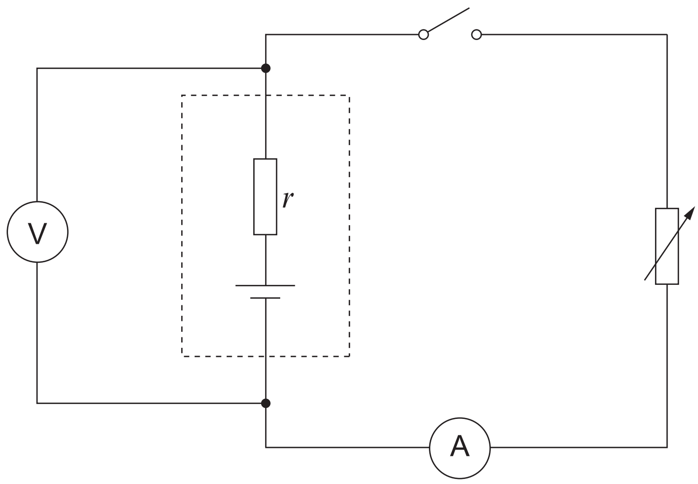
The total number of marks available for this paper is 80.
The number of marks is given in brackets at the end of each question or part-question.
The assessment of the quality of extended response (QER) will take place in question **3**.

PJ*(S20-2420U20-1)

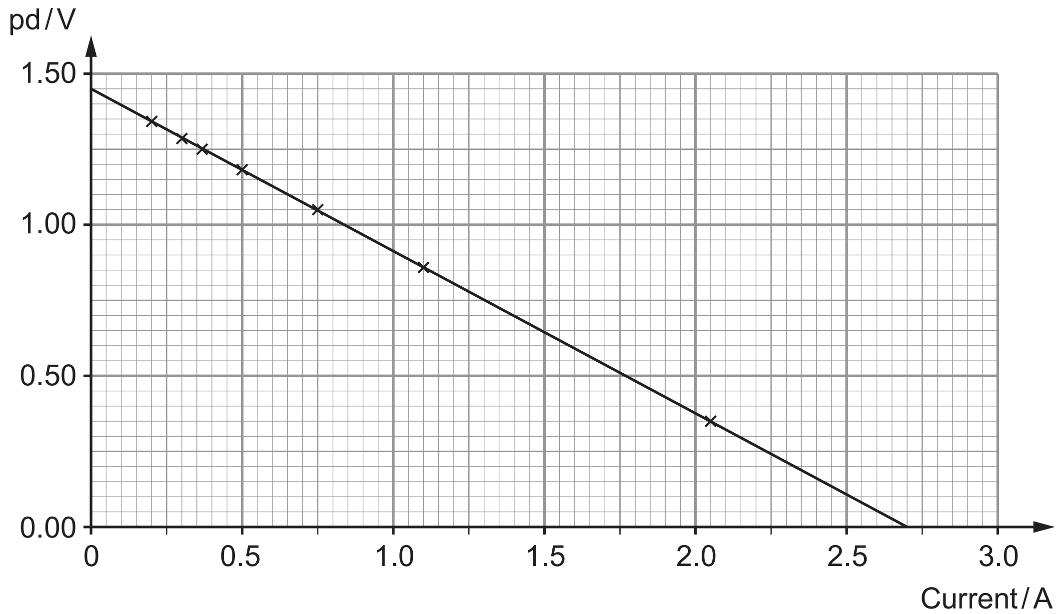
Examiner only

Answer **all** questions.

1. Anwen sets up the circuit shown in order to investigate a cell. The variable resistor is adjusted and readings are obtained from the voltmeter and ammeter.



The readings are plotted as shown.



(a) The manufacturer claims that the emf of the cell is 1.50 V. Explain, in terms of energy, what this statement means. [2]

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Examiner only

(b) An equation applying to a cell of emf, E, and internal resistance, r, is:

$$V = E - Ir$$

(i) Explain how Anwen's graph is in good agreement with this equation. [2]

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(ii) Calculate the gradient of the line and hence determine a value for the cell's internal resistance. [2]

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(iii) Evaluate the manufacturer's claim that the emf is 1.50 V. [1]

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(c) (i) Use the graph to determine the greatest current the cell can supply. [1]

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(ii) State the resistance of the variable resistor when the current is at its maximum. [1]

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Examiner only

2. The current, I, in a wire of cross-sectional area, A, is given by the equation:

$$I = nAve$$

(a) (i) State what the letter n represents in this equation. [1]

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(ii) Using a labelled diagram, derive the equation above. [4]

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(iii) In a physics experiment there is a current of 1.8 A in a nichrome wire of diameter 0.19 mm. For nichrome, $n = 9.0 \times 10^{28}\,\text{m}^{-3}$. Calculate the drift velocity. [2]

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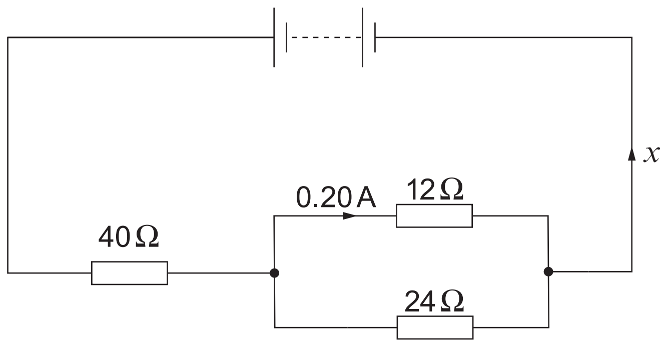
Examiner only

(b) The nichrome wire is cut into different lengths to make three resistors. The resistors are connected in the circuit as shown. The battery has negligible internal resistance.



(i) Explain, in clear steps, why the current, x, is 0.30 A. [3]

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(ii) Calculate the pd across the supply. [3]

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(iii) Show that the power dissipated by the 12 Ω and 24 Ω parallel arrangement is **one fifth** that dissipated by the 40 Ω resistor. [3]

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Examiner only

3. Explain what is meant by a *progressive wave*, **and** explain the difference between a *longitudinal* **and** a *transverse* progressive wave, giving **one** example of each of these types of wave. [6 QER]

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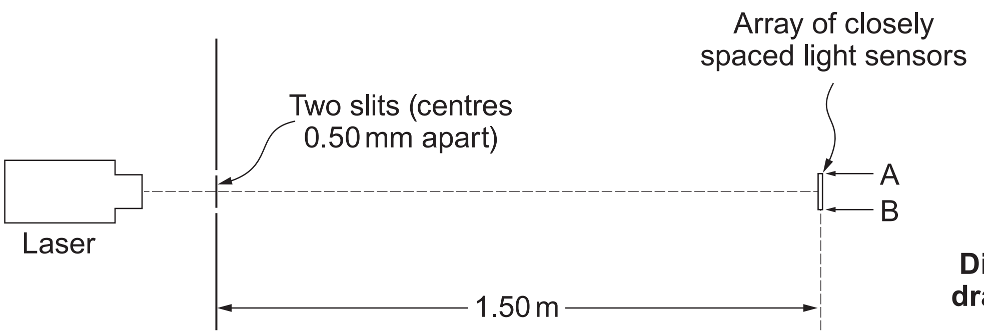
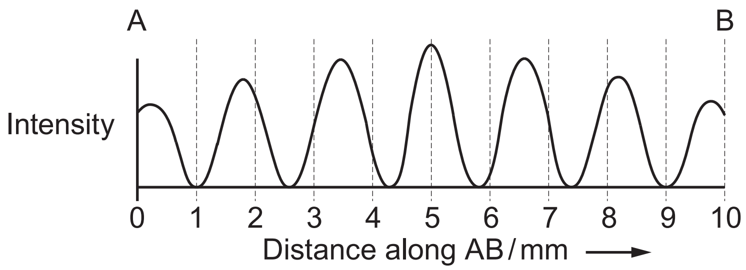
Examiner only

4. (a) A modern version of Young's double slit experiment is set up as shown.



Diagrams not drawn to scale

The array of light sensors is connected to circuitry that displays a graph of light intensity along the line AB.



(i) Calculate a value for the wavelength of light from the laser. [3]

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(ii) Each slit is several wavelengths wide. Suggest why the intensity of the bright fringes decreases the further the fringes are from the centre of the pattern. [2]

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Examiner only

(iii) A student suggests that replacing the laser with one that emits near infra-red (that is infra-red just beyond the end of the visible spectrum) will increase the fringe separation.

I. Explain whether or not the student is right. [2]

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II. State **one** way in which the fringe separation could be increased without changing the laser. [1]

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(b) More than 200 years ago Thomas Young drew a conclusion from his 'fringes' experiment. It was not generally accepted for several years. State what conclusion Young drew, **and** suggest what needed to be done by the scientific community for the conclusion to be accepted. [3]

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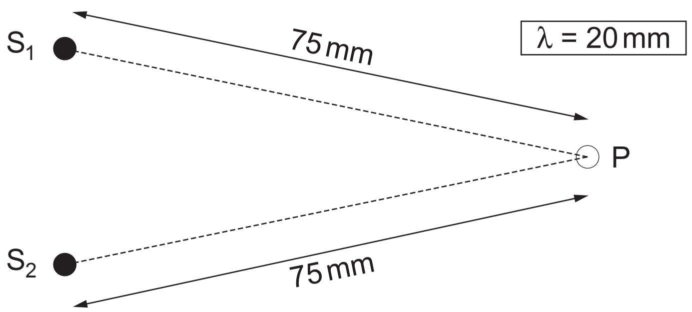
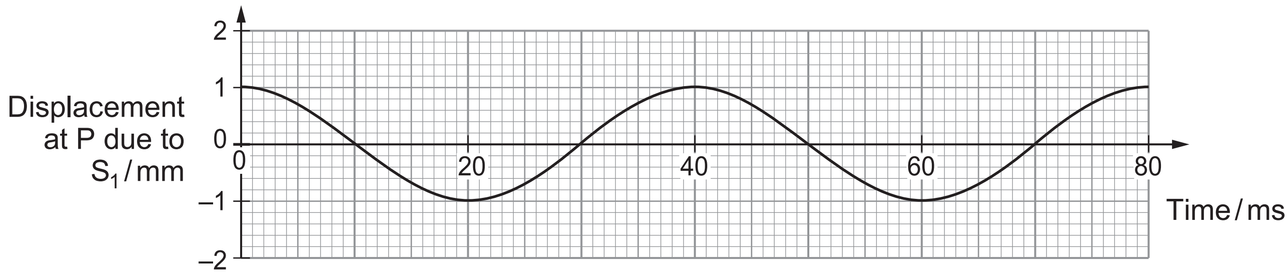
Examiner only

5. (a) The diagram shows where two metal spheres, S_1 and S_2, touch the surface of water (in a tray).

VIEW FROM ABOVE



S_2 remains stationary while S_1 is made to vibrate up and down, so waves spread outwards from it across the water. The wavelength of the waves is 20 mm. A displacement-time graph is given for point P.



(i) Calculate the wave speed. [2]

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(ii) S_1 and S_2 are now made to vibrate **in phase**, to act as wave sources of equal amplitude. Carefully sketch, on the lower two grids **opposite**, displacement-time graphs for point P, due to:

I. waves from S_2; [1]

II. the superposition of waves from S_1 and S_2. [1]

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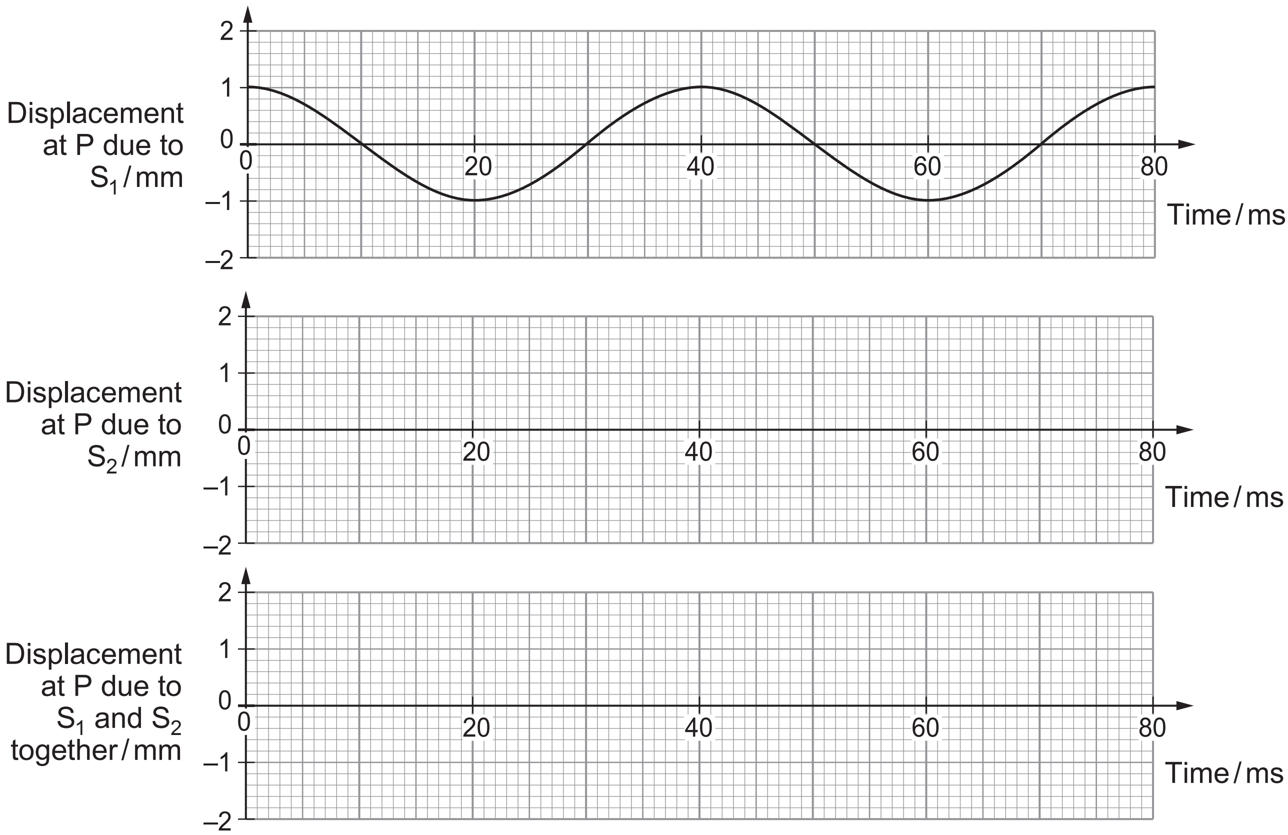
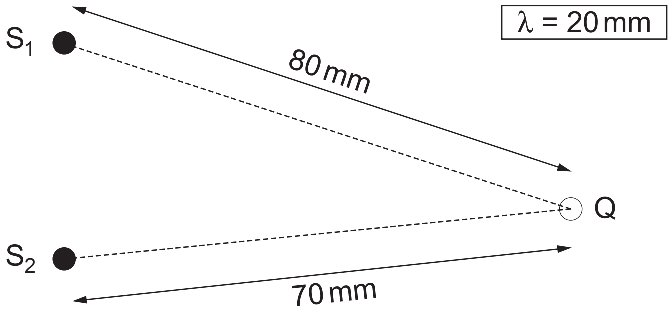


(iii) A student suggests that, with S_1 and S_2 vibrating as before, the displacement at Q (see diagram below) will be zero all the time.

VIEW FROM ABOVE



Evaluate whether or not she is correct, explaining your reasoning. [3]

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Examiner only

(b) Light of wavelength 590 nm is incident normally on a plane diffraction grating. The centres of the grating's slits are 2 000 nm apart.

(i) Calculate the angle to the normal at which **third** order beams emerge. [2]

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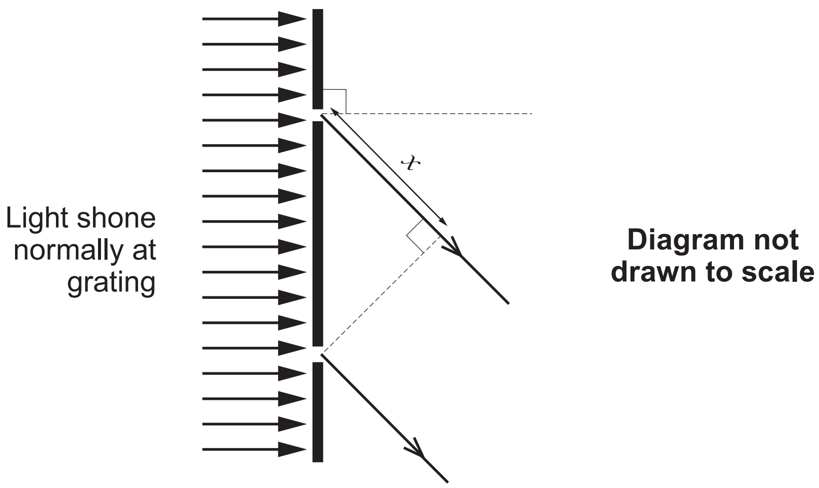
(ii) The diagram shows light emerging **at this angle** from two adjacent slits in the grating.



State the value of x. [1]

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Examiner only

6. (a) The work function of the metal rubidium is 3.62×10^{-19} J. Explain what this statement means. [2]

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(b) The minimum frequency of light that will eject electrons from a surface of work function ϕ is $\dfrac{\phi}{h}$. Explain in terms of photons why this is so. [3]

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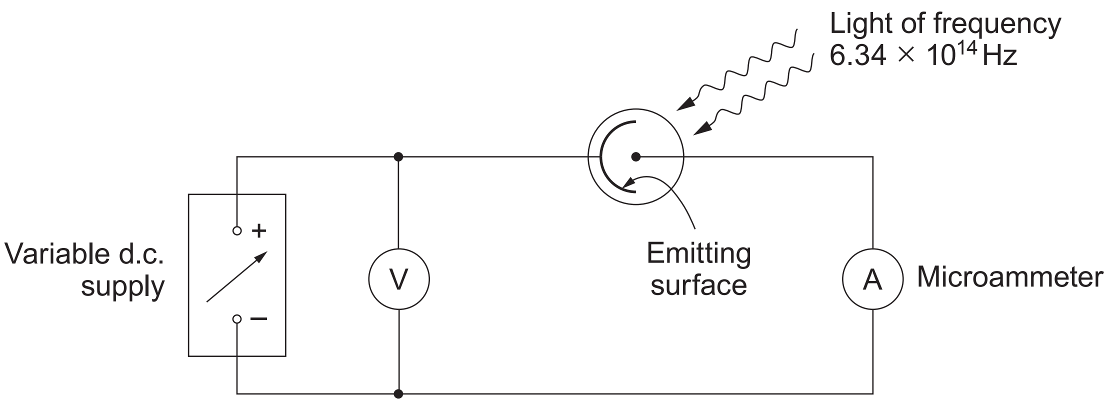
(c) A student has the task of determining whether or not the emitting electrode in a vacuum photocell might be made of rubidium. A light source of known frequency (6.34×10^{14} Hz) is provided. The student sets up the circuit shown.



The student shines the light on the emitting electrode and increases the pd from zero until the current has just fallen to zero. She records this pd (*the stopping pd*) and repeats the procedure, obtaining these results.

Stopping pd / V	0.381	0.366	0.388	0.373	0.371	0.380

Examiner only

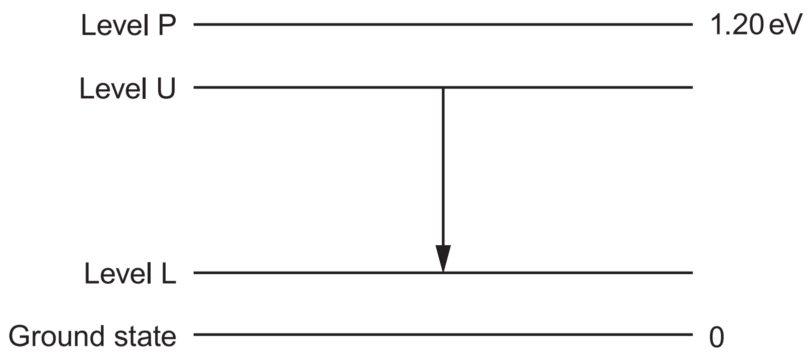
(i) Calculate the mean value and the **absolute** uncertainty of the stopping pd, giving your answers to an appropriate number of significant figures. [3]

(ii) Evaluate whether or not the student should conclude that the emitting electrode might be made of rubidium. [Work function of rubidium = 3.62×10^{-19} J.] [4]

12

Examiner only

7. A simplified energy level diagram for the amplifying medium of a 4-level laser is shown below. The lasing transition occurs between level U and level L.



(a) (i) The laser is pumped to create a population inversion between level U and level L. State what is meant by a population inversion for **this** laser. [1]

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(ii) **Draw three arrows** on the diagram to show the transitions required for this population inversion to be sustained. [1]

(iii) Explain why a population inversion is needed for light amplification to take place. [3]

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(b) Determine whether or not visible light will be produced by the lasing transition from level U to level L, giving your reasoning. [3]

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Examiner only

8. (a) (i) Explain how *multimode dispersion* arises in a glass fibre. [2]

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(ii) State what feature of a *monomode* fibre prevents multimode dispersion. [1]

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(b)

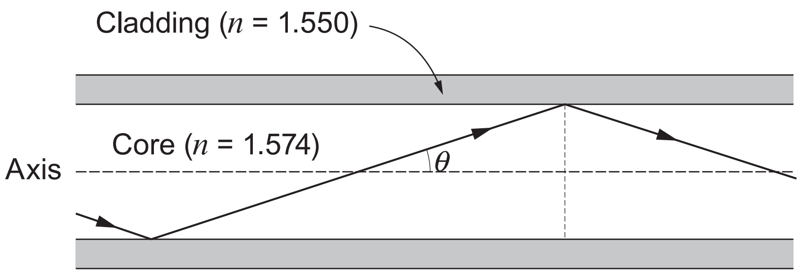

Diagram not drawn to scale

(i) Calculate the largest angle, θ, to the axis (see diagram) at which light can travel for long distances through the core of a multimode fibre, if the refractive index of the core is 1.574, and that of the cladding is 1.550. [3]

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Examiner only

(ii) Explain what would happen to light entering the fibre at an angle to the axis greater than the angle calculated in (b)(i). [2]

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END OF PAPER

Unit 2 2023.pdf

Surname	Centre Number	Candidate Number
First name(s)		2

GCE AS/A LEVEL

2420U20-1

S23-2420U20-1

WEDNESDAY, 24 MAY 2023 – AFTERNOON

PHYSICS – AS unit 2
Electricity and Light

1 hour 30 minutes

For Examiner's use only		
Question	**Maximum Mark**	**Mark Awarded**
1.	**13**	
2.	**11**	
3.	**12**	
4.	**10**	
5.	**10**	
6.	**8**	
7.	**7**	
8.	**9**	
Total	**80**	

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ADDITIONAL MATERIALS

In addition to this paper you will require a calculator and a **Data Booklet**.

INSTRUCTIONS TO CANDIDATES

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INFORMATION FOR CANDIDATES

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The assessment of the quality of extended response (QER) will take place in question **3**(b).

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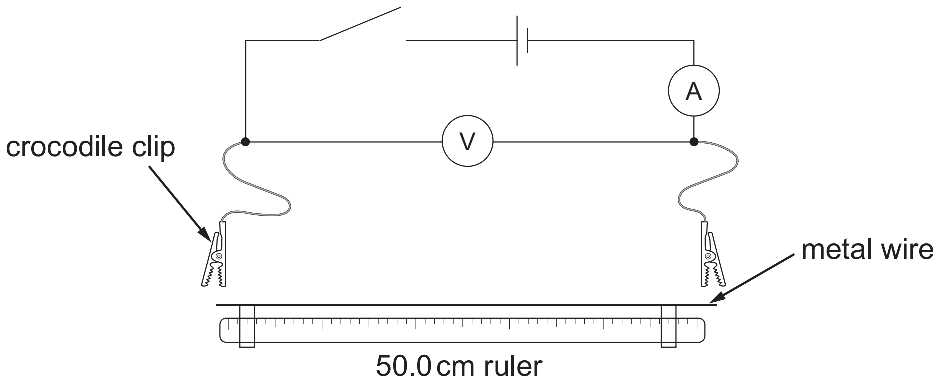
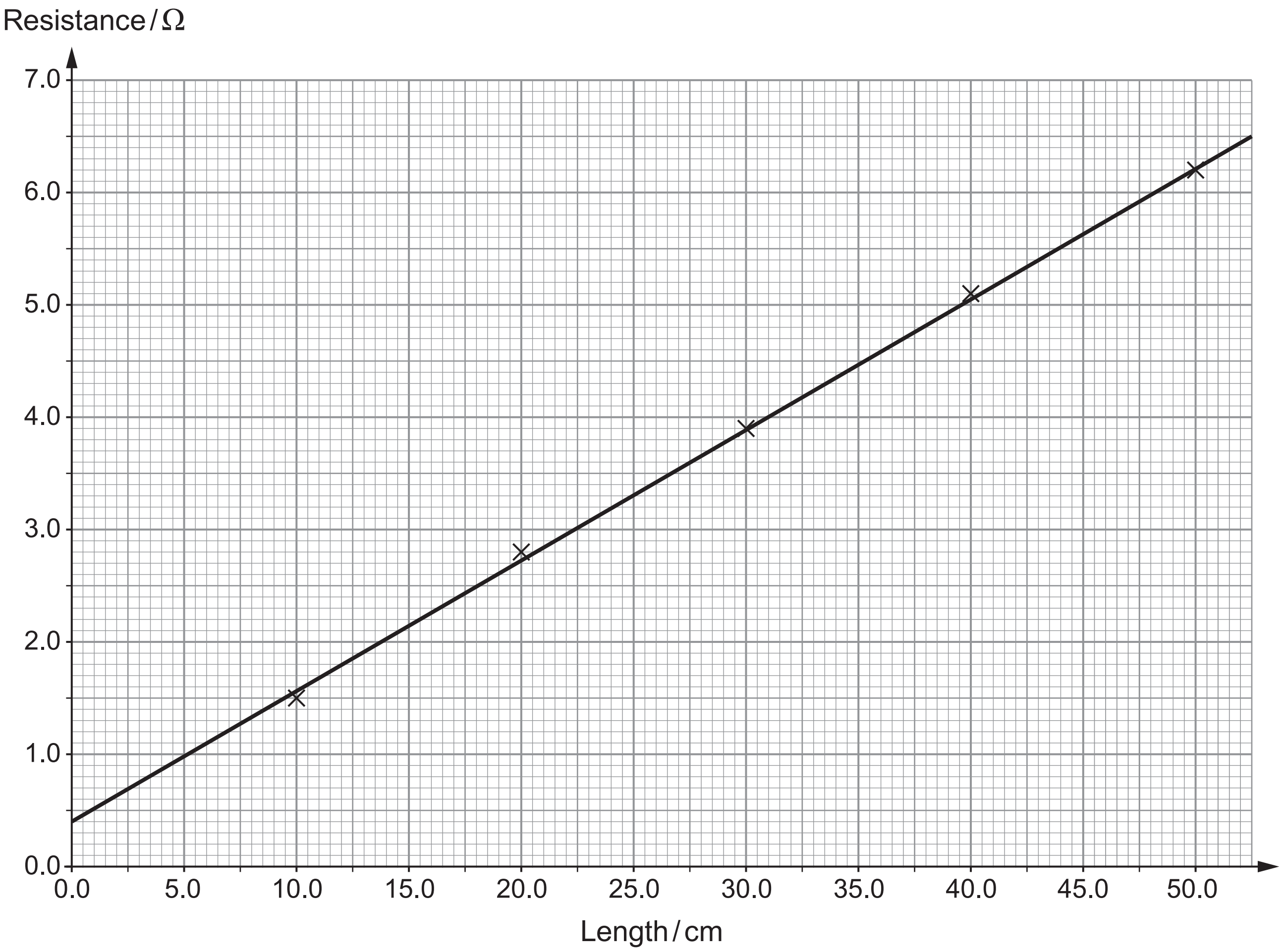
Answer **all** questions.

1. (a) Emily sets up the following circuit in order to investigate how the resistance of a metal wire varies with length.



Emily connects one crocodile clip on the wire at the 0.0 cm mark while the other clip is moved along at suitable intervals to cover the whole range of the wire. She obtains results as shown in the graph below.



Examiner only

(i) Use the graph to obtain a value for the resistance at 0.0 cm and suggest the cause of this resistance. [2]

(ii) Discuss to what extent Emily's results confirm that the variation of resistance of the wire with length is consistent with the equation: [3]

$$R = \frac{\rho l}{A}$$

(iii) The wire used in the experiment has a mean **diameter** of 0.23 mm. Name a measuring instrument that could have been used to take this reading **and** state its likely resolution. [2]

(iv) Use the graph and the mean diameter to determine the resistivity of the metal of the wire. [4]

2420U201 03

Examiner only

(b) Emily used a 1.5 V cell in her experiment as she was concerned about the size of the electrical current in the wire. Suggest her reasons for using the 1.5 V cell. [2]

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Examiner only

2. (a) Define the emf of a cell. [2]

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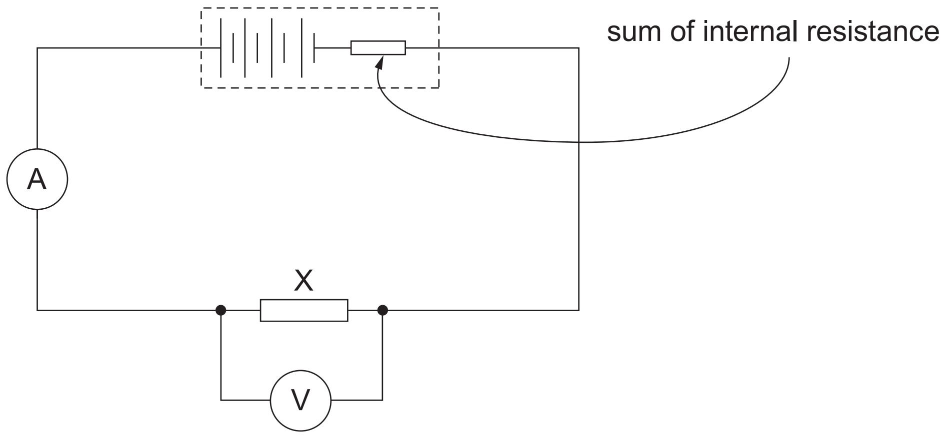
(b) Four identical cells each having an emf of 1.50 V are connected in series to form a 6.00 V battery. The battery is connected across resistor X as shown.



The reading on the ammeter is 0.30 A and the reading on the voltmeter is 5.40 V.

(i) Show that the resistance of resistor X is 18 Ω. [1]

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(ii) Determine the value of the internal resistance of **each individual cell**. [3]

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Examiner only

(iii) Calculate the total power dissipated in the internal resistance of the 6.00 V battery. [2]

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(iv) Resistor X is now replaced with a different resistor, Y. Resistor Y has half the resistance of resistor X. Seren states 'as the resistance is halved, the power dissipated in the internal resistance of the battery will be twice that with resistor X.' Evaluate Seren's claim. [3]

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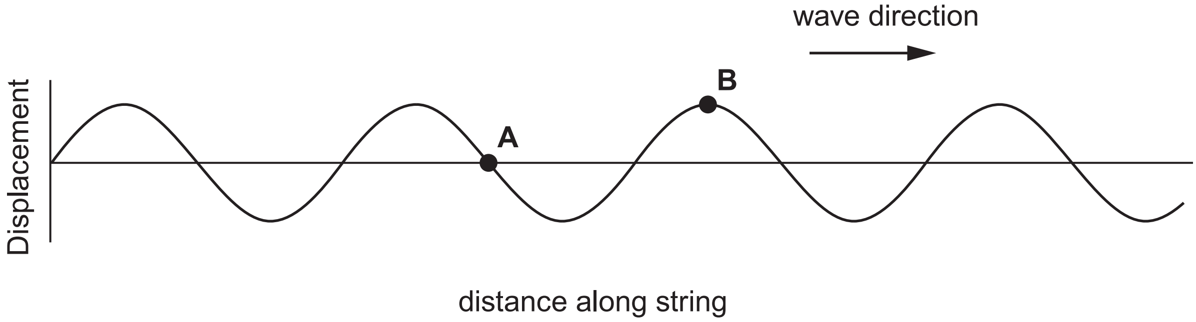
Examiner only

3. (a) A vibration generator attached to one end of a string produces a progressive wave on the string as shown in the diagram (for time $t = 0$). The wavelength of the waves is 15.0 cm.



(i) State what is meant by the amplitude of a progressive wave. [1]

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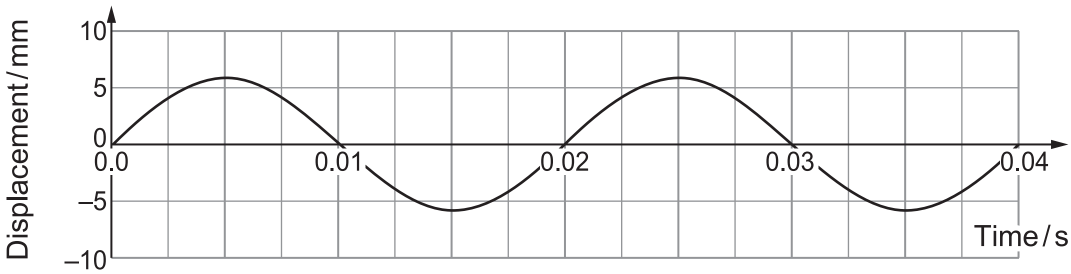
(ii) The following graph shows the displacement of point **A** with respect to time.



Determine the speed of the wave. [3]

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(iii) Sketch the displacement-time graph of point B **on the grid in** (a)(ii). [2]

Examiner only

(b) The progressive wave set-up in part (a) can be modified in order to create a stationary wave pattern. Explain how a stationary wave is formed **and** describe the similarities and differences between a stationary wave and a progressive wave. [6 QER]

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Examiner only

4. Some students investigate the superposition of sound waves. Two identical loudspeakers (S_1 and S_2) are connected to the same signal generator and act as coherent sources.

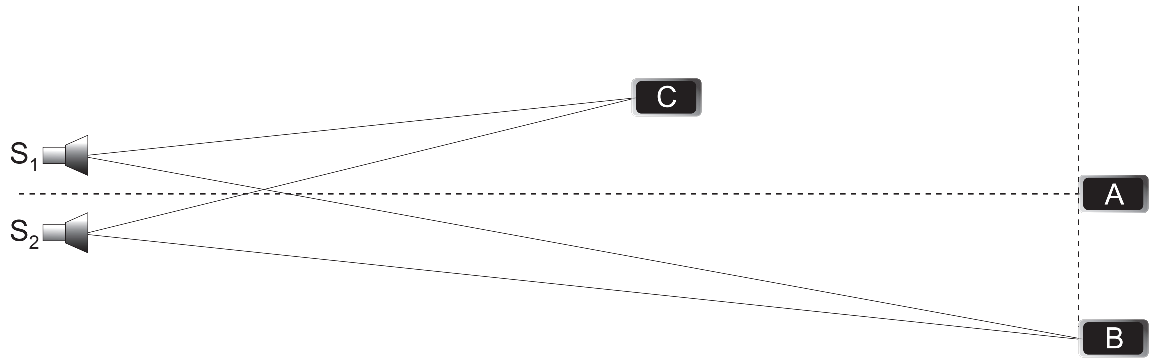

Diagram not drawn to scale

The students use an app on their mobile phones to measure the intensity of the sound signal at different positions in the classroom. Mobile phone A detects a maximum of intensity on the central line. Mobile phone B detects the **first** maximum away from the central maximum.

(a) Explain what is meant by coherent sources. [1]

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(b) (i) The distance S_1B is 8.0 m and the distance S_2B is 7.4 m. State what is meant by the term path difference **and** hence explain why the wavelength of the sound wave must be 0.60 m. [2]

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(ii) Mobile phone C detects a minimum intensity. Determine **two** possible values of the distance ($S_2C - S_1C$). [2]

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Examiner only

(c) (i) State the principle of superposition. [2]

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(ii) Loudspeaker S_2 develops a fault whose only effect is to reduce the amplitude of the sound waves it emits. Explain why the intensities at A and B decrease **and** the intensity at C increases. [3]

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Examiner only

5. (a) Define the refractive index of a material. [1]

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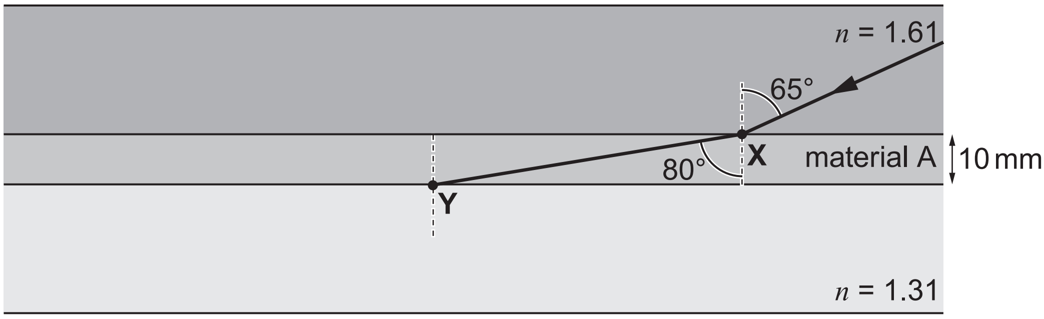
(b) Material A of unknown refractive index, n, is sandwiched between two rectangular blocks of different refractive indices. A ray of light travels from the top block into material A as shown.



(i) Show that the refractive index of material A is approximately 1.5. [2]

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(ii) Determine the time taken for the ray of light to travel through material A from X to Y. [3]

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Examiner only

(c) (i) Confirm that total internal reflection will occur at Y. [3]

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(ii) **On the diagram opposite** continue the path of the light ray inside material A. [1]

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Examiner only

6. (a) State what is meant by the work function, ϕ , of a metal surface. [1]

(b) Sodium will undergo the photoelectric effect when illuminated by visible light, but zinc requires ultraviolet radiation. **Explain** which material has the greater work function. [1]

(c) (i) Use Einstein's photoelectric effect equation to show that the maximum wavelength, λ_{max}, for emission is given by the equation: [2]

$$\lambda_{\text{max}} = \frac{hc}{\phi}$$

(ii) A mixture of red ($\lambda = 650\,\text{nm}$), green ($\lambda = 550\,\text{nm}$) and blue ($\lambda = 450\,\text{nm}$) light is incident on a metal surface of work function of 3.7×10^{-19} J. Determine which wavelength or wavelengths of light will be **unable** to release electrons from the metal surface. [2]

(iii) Explain, in terms of photons, whether or not the intensity of the light will affect the maximum kinetic energy of the released electrons. [2]

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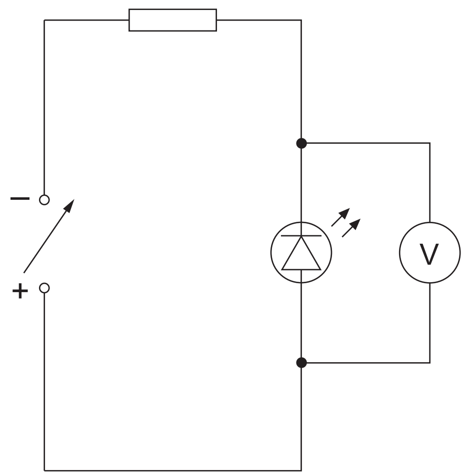
Examiner only

7. (a) Lewis uses the following circuit to find the minimum pd, V_{min}, across an LED at which light is emitted by the diode. He collects his data in a dark room where he varies the pd of the power supply until the LED lights.



Lewis measures the minimum pd, V_{min}, five times and his results are shown below.

V_{min}/V	1.81	1.87	1.93	1.84	1.90

(i) Determine the mean value for V_{min} along with its **percentage** uncertainty. [3]

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Examiner only

(ii) Lewis uses the following equation to calculate the wavelength of the light expected to be produced by the LED:

$$eV_{\text{min}} = \frac{hc}{\lambda}$$

Calculate the mean value for λ and its **absolute** uncertainty. [3]

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(b) Lewis noted the value for V_{min} when he noticed that the LED had turned on. Suggest an improvement to his method. [1]

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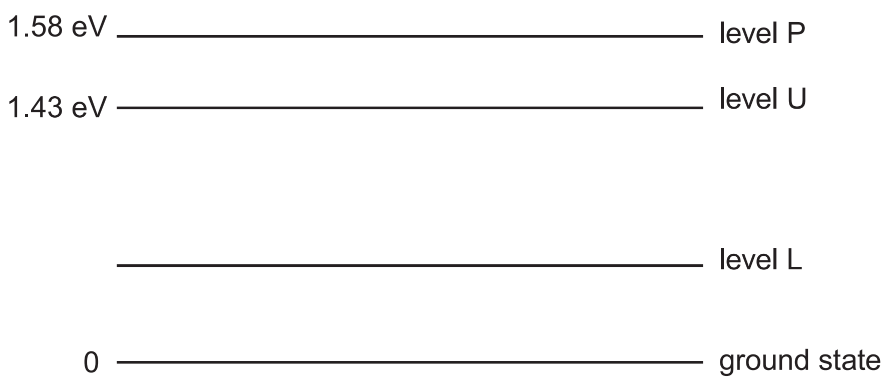
Examiner only

8. The following diagram shows the energy levels for the amplifying medium in a 4-level gas laser.



(a) The laser transition between level U and level L produces stimulated photons of wavelength 1.06×10^{-6} m. Determine the value of energy level L in **eV**. [3]

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(b) When explaining the operation of this laser, a student writes the following:

Pumping is required to move electrons from ground level to level P which is a metastable state. This creates a population inversion between P and ground which is needed for the laser to work.

State **two** mistakes in the student's explanation. [2]

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Examiner only

(c) Semiconductor lasers are regularly used in DVD players and bar code scanners. State **two** advantages that these semiconductor lasers have over gas lasers. [2]

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(d) As electronic technology advances, outdated electronic devices and their components are discarded. It has been estimated that 50 million tonnes of electronic waste is produced each year and roughly 80% of this is finding its way into landfill. Discuss the effects this electronic waste may have on society. [2]

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END OF PAPER